Online supplement to “Neural means and kernel corrections for operator learning”

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Abstract

This supplement gives the conditional finite-sample bounds, decision rules, conformal coverage results, and proofs supporting the main article. It also describes the additional benchmark analyses, implementation and computational costs, numerical evaluations, and sensitivity experiments. Sections and results prefixed by S refer to this document; other numbered references refer to the main article.

S1 Guarantees for the fitted stages

S1.1 Full-map symmetry averaging does not increase the risk

The elasticity problem behind the benchmark is invariant under the reflection $x_1 \mapsto 1 - x_1$: the domain, the boundary partition, and the constitutive model are all symmetric, the input distribution has a reflection-invariant covariance, and reflecting the load reflects the stress field. On the discrete grid this is expressed by two permutation matrices: $S \in \mathbb{R}^{p\times p}$ reverses the load samples and $T \in \mathbb{R}^{q\times q}$ reverses the rows of the stress field. Both are orthogonal involutions. Supplement S4.2 examines the discrete equivariance relation $G(Su) = T\,G(u)$ empirically. The following proposition assumes this relation exactly.

Proposition S1.1 (symmetrization)**.** *Assume $G(Su) = TG(u)$ for μ-a.e. u, that μ is invariant under S, and that T is an orthogonal involution ($T^2 = I$). For a measurable $\widehat{G} : \mathcal{U} \to \mathcal{V}$ define its symmetrization*

$$\widehat{G}_S(u) \;=\; \tfrac{1}{2}\left(\widehat{G}(u) + T\,\widehat{G}(Su)\right).$$

Then, for every loss $\ell(\cdot, v)$ that is convex in its first argument and satisfies $\ell(Tw, Tv) = \ell(w, v)$,

$$\mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}(u), G(u)\big).$$

Proof. See Supplement S8.1.

The relative error satisfies both hypotheses when the target norm is nonzero. The proposition guarantees that test-time reflection averaging cannot increase population risk under exact equivariance and distributional invariance. For a finite training sample, Jensen’s inequality also shows that the reflection-augmented empirical loss upper-bounds the empirical loss of the symmetrized predictor. The objectives are generally different, so the proposition does not guarantee that training with augmentation improves the fitted model. The near-reflection comparisons are consistent with symmetry of the distributed solver output. In the experiment, the averages improve twenty-nine of thirty members across the two MLP variants and the refiner, and worsen one by a thousandth of a point. For the refiner, equivariance of its conditioning field is a sufficient additional condition for this guarantee. The convexity argument is standard; Lyle et al. (2020) give a general account.

A supplied conditioning field is an additional input. For a fitted refiner $F(u, h)$ and a fixed kernel predictor $h(u)$, define $f(u) = F(u, h(u))$. The proposition concerns

$$f_S(u) = \tfrac{1}{2}\big(F(u, h(u)) + TF(Su, h(Su))\big).$$

The implementation instead uses $Th(u)$ as the second branch's conditioning field. Thus $h(Su) = Th(u)$ is a sufficient additional condition for identifying the implemented average with f_S. Equivariance of the target and invariance of the load distribution do not by themselves give this property to a fitted kernel predictor, and training the refiner with randomly reflected triples does not force exact equivariance; the non-increase guarantee for its averaging operation is conditional on that identity.

The effect of this difference can be bounded. Write $b(u) = \frac{1}{2}(F(u, h(u)) + TF(Su, Th(u)))$ for the implemented average and $R(g) = \mathbb{E}\|g(u) - G(u)\|_2 / \|G(u)\|_2$, assuming nonzero targets almost surely and finite quantities below. Under the proposition's exact symmetry assumptions,

$$\begin{aligned} R(b) &\le R(f_S) + \frac{1}{2}\mathbb{E}\frac{\|F(Su, Th(u)) - F(Su, h(Su))\|_2}{\|G(u)\|_2} \\ &\le R(f) + \frac{1}{2}\mathbb{E}\frac{\|F(Su, Th(u)) - F(Su, h(Su))\|_2}{\|G(u)\|_2}. \end{aligned} \tag{S1.1}$$

Indeed, the triangle inequality bounds the risk difference by $\mathbb{E}\|b - f_S\|_2/\|G\|_2$, and orthogonality of T gives the displayed defect exactly; the second inequality is Proposition S1.1. If $F(u, \cdot)$ is uniformly L_h-Lipschitz, the additional term is at most $(L_h/2)\mathbb{E}[\|Th(u) - h(Su)\|_2/\|G(u)\|_2]$. This defect and the Lipschitz constant are not measured in the experiments, so the inequality is not evaluated numerically.

The two averages can have different risks even for an exactly symmetric target. Let the input take the values $+1$ and -1 with equal probability, let S exchange them, and let T swap two output coordinates. Set $G(u) = (1, 1)$, $h(+1) = (1, 1)$ and $h(-1) = (3, 3)$. Define $F(+1, z) = z$ and $F(-1, z) = (4, 4) - z$. The four evaluations are

$$\begin{aligned} &F(+1, h(+1)) = (1, 1), \quad F(-1, Th(+1)) = (3, 3), \\ &F(-1, h(-1)) = (1, 1), \quad F(+1, Th(-1)) = (3, 3). \end{aligned}$$

Thus $f(u) = G(u)$ at both inputs, and the full-map average also has zero error. Averaging with $Th(u)$ instead gives $(2, 2)$ at both inputs, with relative error one. This finite example disproves a general guarantee for the supplied-field operation and attains equality in (S1.1) (here $L_h = 1$).

S1.2 The kernel-flow regularizer and leave-half-out error

The kernel-flow (KF) loss of Owhadi and Yoo (Owhadi and Yoo, 2019), in the ℓ^2 variant used by Yoo and Owhadi (2021) to regularize deep networks, defines the optional regularizer in the ablation study. Let $z_i = \phi_\theta(u_i)$ be the features of a batch $B = \{1, \dots, b\}$ (b even), let k be a positive definite kernel on feature space, and for $C \subset B$, $|C| = b/2$, let I_C denote the k-interpolant of $\{(z_j, v_j) : j \in C\}$. The batch KF loss is

$$e_2(\theta; C) \;=\; \sum_{i \in B} \big\|v_i - I_C(z_i)\big\|_2^2. \tag{S1.2}$$

Proposition S1.2 (KF loss is a leave-half-out estimate)**.** *Let C be drawn uniformly among the subsets of B of size $b/2$ and assume k restricted to $\{z_i\}_{i\in B}$ is strictly positive definite. Then $e_2(\theta; C) = \sum_{i \in B\setminus C} \|v_i - I_C(z_i)\|_2^2$, and*

$$\mathbb{E}_C\big[e_2(\theta; C)\big] \;=\; \frac{b}{2}\, \mathbb{E}_{C,\, i\sim \mathrm{Unif}(B\setminus C)}\big[\big\|v_i - I_C(z_i)\big\|_2^2\big],$$

i.e. $\frac{2}{b}\, e_2$ is an unbiased estimator, over the split randomness, of the expected squared error of the kernel predictor on a held-out point when trained on a random half of the batch. If the parameterized feature and kernel maps are differentiable and the derivative of $e_2(\theta; C)$ is bounded in a parameter neighborhood by a random variable integrable over (B, C), then, when (B, C) are resampled at every step, the stochastic gradient $\nabla_\theta e_2(\theta; C)$ is unbiased for $\nabla_\theta\, \mathbb{E}_{B,C}[e_2(\theta; C)]$.

Proof. See Supplement S8.2.

The implemented ablation uses normalized targets and the mean squared loss over all bq batch-output entries, with predictor $K_{BC}(K_{CC} + 10^{-3}I)^{-1}Y_C$. Its errors on C need not vanish: on those rows the residual is $10^{-3}(K_{CC} + 10^{-3}I)^{-1}Y_C$. Consequently the exact leave-half-out identity above does not apply to the regularized loss. The learned Gaussian bandwidth multiplier is combined with a batch median distance computed under `no_grad`; the resulting gradient also omits the dependence of that median on the features. This stop-gradient rule differs from the full derivative in Proposition S1.2. Neither the ideal identity nor its regularized analogue estimates the population risk of the adaptively trained network.

S1.3 Capacity of stacking with fixed members

Convexity bounds the loss of a convex combination by the corresponding weighted average of member losses. A separate uniform-deviation argument bounds the cost of fitting the weights when the candidate members are fixed independently of an i.i.d. validation sample and the losses have a population bound.

Proposition S1.3 (validation stacking)**.** *Let $f_1, \dots, f_M : \mathcal{U} \to \mathcal{V}$ be fixed maps, let $\Delta = \{w \in \mathbb{R}^M : w_m \ge 0, \sum_m w_m = 1\}$, and for $w \in \Delta$ write $F_w = \sum_m w_m f_m$. Assume the members' per-sample relative errors are bounded: $\ell(f_m(u), G(u)) \le B$ for μ-a.e. u and every m.*

(i) For every $w \in \Delta$, $\ell(F_w(u), G(u)) \le \sum_m w_m\, \ell(f_m(u), G(u))$ pointwise; in particular $R(F_w) \le \sum_m w_m R(f_m) \le \max_m R(f_m)$, and $\ell(F_w(u), G(u)) \le B$.

(ii) Let $\widehat{w}$ minimize the empirical risk $\widehat{R}(w) = \frac{1}{m}\sum_{i=1}^{m} \ell(F_w(\tilde{u}_i), G(\tilde{u}_i))$ over Δ, computed on an i.i.d. validation sample $\tilde{u}_1, \ldots, \tilde{u}_m \sim \mu$ independent of $f_1, \ldots, f_M$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$R(F_{\widehat{w}}) \;\le\; \min_{w\in\Delta} R(F_w) \;+\; \frac{8B}{m} \;+\; B\sqrt{\frac{2\big((M-1)\log\big(m(M-1)+1\big) + \log(2/\delta)\big)}{m}}.$$

Proof. See Supplement S8.3.

With $M = 5$, $m = 1000$ and $\delta = 0.05$, the excess term is about $0.28B$. Substituting $B = 0.25$, the largest observed member error, gives roughly seven percentage points, if that value is a uniform population bound. This excess term is much larger than the measured selection effects. The result supplies the scaling $\sqrt{M \log m/m}$; the much smaller validation-to-test gaps are empirical measurements.

The stack the pipeline deploys on the structural-mechanics benchmark is more general than a single convex vector: the weights are affine and vary per grid point, fitted by ridge on the validation split, and selection between the affine and global convex stacks uses a half-split comparison: each is fitted on one half, evaluated on the other, and the one with lower error is refitted on the full split. Proposition S1.3 does not cover that construction. The following two statements give conditional capacity bounds for related fixed-candidate selection and affine classes.

Lemma S1.4 (validated selection)**.** *Let $g_1, \ldots, g_K$ be fixed maps from $\mathcal{U}$ to $\mathcal{V}$. Assume $\ell(g_k(u), G(u)) \le B$ for μ-a.e. u and every k, and let $\widehat{k}$ minimize the empirical risk over an i.i.d. sample of size m' drawn from μ independently of everything used to build the g_k. Then, with probability at least $1 - \delta$,*

$$R(g_{\widehat{k}}) \;\le\; \min_{k\le K} R(g_k) \;+\; B\sqrt{\frac{2\log(2K/\delta)}{m'}}.$$

Proof. See Supplement S8.4.

The half-split comparison uses $K = 2$ and $m' = 500$. The lemma applies only if both candidate stacks were constructed independently of the evaluation half. Refitting the selected stack changes the predictor. The following proposition treats affine fitting under independence and boundedness assumptions.

Proposition S1.5 (per-pixel affine stacking)**.** *Fix members $f_1, \ldots, f_M$, write $D = \sqrt{M+1}$, and for each output coordinate j let $\varphi_j(u) = (f_1(u)_j, \ldots, f_M(u)_j, b) \in \mathbb{R}^{M+1}$, where b bounds all member and target values: $|G(u)_j| \le b$, $|f_m(u)_j| \le b$ for μ-a.e. u and all j, m. Consider the per-coordinate affine class $\{u \mapsto w^\top \varphi_j(u) : \|w\|_2 \le W\}$ and the coordinate risks $R_j(w) = \mathbb{E}\,(w^\top \varphi_j(u) - G(u)_j)^2$, estimated on an i.i.d. validation sample of size m independent of the members.*

(i) If $\widehat{w}_j$ minimizes the empirical risk of coordinate j over the class, then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_{j=1}^{q}\Big[R_j(\widehat{w}_j) - \min_{\|w\|_2\le W} R_j(w)\Big]$$
$$\le \frac{8\,Wb^2D\,(1+WD) + 2\,b^2(1+WD)^2\sqrt{\tfrac{1}{2}\log(4q/\delta)}}{\sqrt{m}}.$$

(ii) If instead $\widehat{w}_j$ minimizes the ridge objective $\widehat{R}_j(w) + \lambda\|w\|_2^2$ with $\lambda \ge 0$ and satisfies $\|\widehat{w}_j\|_2 \le W$, the same bound holds with an additional λW^2 on the right.

Proof. See Supplement S8.5.

For fixed M, b, W, this is an $m^{-1/2}$ bound on coordinatewise mean-squared risk, with logarithmic dependence on the number of coordinates. It is not a relative-error bound. Its constants depend polynomially on b and W; substituting the measured values makes the bound exceed the realized excess by many orders of magnitude. Those substitutions are illustrative, since a finite observed maximum is not a population bound.

A fixed grid of residual corrections gives a further composite class. For each configuration, correction of training residuals is affine in the stack weights, so the same uniform-deviation argument applies with a transformed feature map. The following result requires the training arrays, members and correction configurations to be fixed independently of validation.

Corollary S1.6 (capacity bound for a fixed composite correction class)**.** *Assume the setting of Proposition S1.5, and let Γ be a finite set of correction configurations, each defining weights $c_\gamma(u) \in \mathbb{R}^{n_{\mathrm{tr}}}$ with the corrected predictor*

$$h_{w,\gamma}(u)_j \;=\; w_j^\top \varphi_j(u) + c_\gamma(u)^\top\big(Y_j - \Phi_j w_j\big),$$

where $Y_j \in \mathbb{R}^{n_{\mathrm{tr}}}$ collects the training targets of coordinate j, and $\Phi_j \in \mathbb{R}^{n_{\mathrm{tr}}\times(M+1)}$ has rows $\varphi_j(u_i)^\top$, including the final column of intercept features equal to b; $\gamma = 0$ (no correction, $c_0 \equiv 0$) is included in Γ. Suppose the correction weights are uniformly bounded, $\sup_{u,\gamma}\|c_\gamma(u)\|_1 \le \kappa$. Assume that Y_j, Φ_j, the members and all c_γ are fixed independently of the validation sample, and that their training target and feature entries satisfy the same absolute bound b. For any validation-measurable choice $(\widehat{w}, \widehat{\gamma})$ with $\|\widehat{w}_j\|_2 \le W$ and $\widehat{\gamma}$ minimizing the empirical validation risk, aggregated over coordinates, among $\gamma \in \Gamma$ at the fitted $\widehat{w}$ (the grid choice is shared across coordinates), with probability at least $1-\delta$,

$$\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\widehat{\gamma}}\big) \le \min_{\gamma\in\Gamma}\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\gamma}\big)$$
$$+\frac{2(1+\kappa)^2 b^2(1+WD)}{\sqrt{m}}\left(4WD + (1+WD)\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2}}\right).$$

Since $\gamma = 0$ is among the candidates, the oracle on the right is at most the fitted stack's risk, and combining with Proposition S1.5(ii) bounds the composite risk against the best affine stack, provided the hypotheses of both results hold (with failure probabilities allocated between them). Here b and W must be population and class bounds; the observed values are diagnostics of them. For kernel ridge weights, κ is supplied by the next lemma.

Proof. See Supplement S8.7.

For kernel ridge regression, the correction-weight bound follows analytically. In the pipeline each $\gamma = (s, \lambda)$ is a length scale and a nugget, and the weights are those of kernel ridge regression, $c_\gamma(u) = (K_\gamma + n\lambda I)^{-1} k_\gamma(X, u)$, so that $\widehat{r}(u) = c_\gamma(u)^\top R$ in the notation of Section 3.3. Their ℓ^1 norm is bounded uniformly in u by the nugget alone.

Lemma S1.7 (uniform bound on kernel ridge weights)**.** *Let k be a positive definite kernel with $k(u,u) \le k_{\max}$ for all u, let $X = (u_1, \dots, u_n)$ be any design with Gram matrix K, and let $\lambda > 0$. The kernel ridge weights $c_\lambda(u) = (K + n\lambda I)^{-1} k(X, u)$ satisfy, for every u,*

$$\|c_\lambda(u)\|_1 \;\le\; \sqrt{n}\,\|c_\lambda(u)\|_2 \;\le\; \frac{1}{2}\sqrt{\frac{k(u,u)}{\lambda}} \;\le\; \frac{1}{2}\sqrt{\frac{k_{\max}}{\lambda}},$$

and the constant $\frac{1}{2}$ cannot be improved: for every $\lambda \in (0,1)$ there are a kernel, a two-point design and a point u at which the first and last members are equal. In particular, for the residual correction of Section 3.3 — a Matérn kernel with $k(u,u) = 1$ and the nugget grid $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$ — the hypothesis of Corollary S1.6 holds, whatever the length scale and the design, with

$$\kappa_{\mathrm{an}} \;:=\; \tfrac{1}{2} \max_{\gamma \in \Gamma} \lambda_\gamma^{-1/2} \;=\; 500.$$

Proof. See Supplement S8.8.

Positive semidefiniteness of the Gram matrix of the design together with u confines $k(X,u)$ to the range of K with $k(X,u)^\top K^{+} k(X,u) \le k(u,u)$, and $\mu/(\mu + n\lambda)^2 \le 1/(4n\lambda)$ for every eigenvalue μ. Thus κ_{an} is a uniform bound determined by the kernel diagonal and nugget grid. The excess term contains $(1+\kappa)^2$; with $\kappa_{\mathrm{an}} = 500$, it exceeds the measured errors by many orders of magnitude. The realized weights have a smaller scale: on a test probe of five hundred points the largest ℓ^1 norm over the grid averages 32.2 over the ten seeds (Section S10), about a fifteenth of κ_{an}. Substituting that value would reduce the excess term by about a factor of 240, but a finite-probe maximum does not supply the required uniform bound.

The corollary bounds coordinatewise mean-squared risk for candidates fixed independently of validation. The deployed estimator instead selects by mean relative L^2 error, tunes its kernel stages on an 8000-row subsample, refits the selected kernel on all 19000 rows, and reuses the validation split for checkpoint and member selection. These operations are outside the corollary's assumptions. Likewise, observed maxima do not establish the population bounds B, b, L, W required by the preceding capacity results.

S2 Decision rules: margins and per-coordinate selection

The two-member criterion is exact for population second moments. Applying it to estimates introduces a separate estimation problem. The following bound assumes conditionally sub-Gaussian estimation error and assigns ties to the decision not to mix.

Proposition S2.1 (margin law for a threshold rule)**.** *Let $D = \varrho - e_1/e_2$ be the gap in Proposition 6.1(i), and define the binary decision $d(x) = \mathbf{1}\{x < 0\}$. Thus $d(D) = 1$ means* *that mixing strictly helps, and $d(0) = 0$. Suppose, conditionally on D, an estimate $\widehat{D}$ satisfies, for some fixed $\sigma > 0$,*

$$\mathbb{E}\Big[\exp\{t(\widehat{D} - D)\} \mid D\Big] \le \exp(t^2\sigma^2/2), \qquad t \in \mathbb{R}.$$

Then:

(i) for $D \neq 0$, writing $\Delta = |D|$, $\mathbb{P}(d(\widehat{D}) \neq d(D) \mid D) \le \exp(-\Delta^2/(2\sigma^2))$;

(ii) for every $\tau \ge 0$, including when $D = 0$,

$$\mathbb{P}\big(d(\widehat{D}) \neq d(D),\ |\widehat{D}| \ge \tau\big) \le \exp(-\tau^2/(2\sigma^2)).$$

Proof Condition on D. If $D < 0$, a wrong decision requires $\widehat{D} \ge 0$, hence $\widehat{D} - D \ge |D|$. If $D > 0$, a wrong decision requires $\widehat{D} < 0$, hence $D - \widehat{D} > |D|$. The one-sided Chernoff bound obtained by minimizing $\exp(t^2\sigma^2/2 - t\Delta)$ over $t > 0$ gives (i). For (ii), when $D < 0$ the event in question implies $\widehat{D} \ge \tau$, so $\widehat{D} - D \ge \tau$. When $D \ge 0$, including a tie, it implies $\widehat{D} < 0$ and $\widehat{D} \le -\tau$, so $D - \widehat{D} \ge \tau$. The same one-sided bound gives $\exp(-\tau^2/(2\sigma^2))$ in either case, and averaging over D proves (ii). At $\tau = 0$ the bound is the trivial upper bound one. ■

Part (ii) bounds the joint event of a wrong decision and a large observed margin. It does not bound the error rate conditional on selecting such a margin; that would require division by $\mathbb{P}(|\widehat{D}| \ge \tau)$. The hypothesis also implies conditional centering of the estimation error and a sub-Gaussian tail bound, neither of which follows from a small pooled mean or standard deviation. The differences across the 840 dependent member pairs describe the observed validation-to-test transfer scale. In particular the OCO-2 validation and test blocks have the distribution shift described in Section 5. The margin-bin accuracies in Section 5.2 describe empirical transfer and do not verify the conditional sub-Gaussian assumption.

Minimizing a separable squared output metric reduces to independent choices for each coordinate. This separability does not extend to the reported means of relative norms.

Proposition S2.2 (per-coordinate combination under diagonal metrics)**.** *For weights $v = (v_{mj})$ that may depend on the output coordinate, let $f_v(u)_j = \sum_m v_{mj} f_m(u)_j$. For any diagonal metric with positive weights w,*

$$\mathbb{E}\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2 = \sum_j w_j^2\, \mathbb{E}\big(f_v(u)_j - G(u)_j\big)^2,$$

so the minimizing weights solve q separate problems, one per output coordinate, none of which involves w. The per-coordinate optimum is the same for every diagonal metric, and it weakly dominates every combination whose weights are shared across coordinates, for all such metrics simultaneously.

Proof. See Supplement S8.14.

Here each coordinate's feasible weights may be chosen independently, and the optimum is a population optimum. With fixed positive diagonal weights, the squared risk separates in this way. An empirical minimizer has the same algebraic separation for the empirical objective, but need not be a population minimizer. A common per-sample weight also preserves separation.

Corollary S2.3 (weighted metrics and the reported error)**.** *Let $a : \mathcal{U} \to [0, \infty)$ be measurable with $\mathbb{E}[a(u)\,\|f_v(u) - G(u)\|_2^2] < \infty$. Then, for every diagonal w,*

$$\mathbb{E}\Big[a(u)\,\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\,\big(f_v(u)_j - G(u)_j\big)^2\Big],$$

so the minimization still splits into q per-coordinate problems, none involving w. Taking $a(u) = \|G(u)\|_2^{-2}$ and $w = \mathbf{1}$ makes the left side the mean squared relative error: the population combination minimizing each weighted coordinate risk is simultaneously optimal, within its class, for the squared relative error and for every diagonal reweighting of it. The reported metric, $\mathbb{E}\,\|\rho_{f_v}\|_2$ with ρ the normalized residual of Section 6.2, is controlled through $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$; the gap between the two sides is a predictor-specific dispersion factor. For the fitted mechanics mixtures it is about 0.94 *on validation; it must be measured separately for other predictors and samples.*

Proof. See Supplement S8.15.

Proposition S2.2 and its corollary describe the population optimum; the pipeline computes an empirical selection, coordinate by coordinate, on the validation split. For candidates fixed independently of an i.i.d. validation sample, the following inequality bounds its finite-sample selection cost.

Proposition S2.4 (empirical per-coordinate selection)**.** *Let $f_1, \dots, f_M$ be fixed maps, and let $a : \mathcal{U} \to [0, A]$ be a bounded per-sample weight. Suppose $a(u)\,(f_m(u)_j - G(u)_j)^2 \le L$ for μ-a.e. u, every member m and coordinate j. Take a fixed independently of validation. On an i.i.d. validation sample from μ of size m_v independent of the members, let $\widehat{m}(j)$ minimize the empirical weighted risk of coordinate j over the M members, and let the combination take coordinate j from member $\widehat{m}(j)$. Then, with probability at least $1 - \delta$, simultaneously for every coordinate,*

$$R_j\big(\widehat{m}(j)\big) \le \min_{m \le M} R_j(m) + 2L\sqrt{\frac{\log(2Mq/\delta)}{2m_v}},$$
$$R_j(m) = \mathbb{E}\big[a(u)\big(f_m(u)_j - G(u)_j\big)^2\big].$$

and consequently, for every diagonal metric w at once,

$$\sum_j w_j^2 R_j(\widehat{m}(j)) \le \sum_j w_j^2 \min_m R_j(m) + 2L\Big(\sum_j w_j^2\Big)\sqrt{\frac{\log(2Mq/\delta)}{2m_v}}.$$

Proof. See Supplement S8.16.

The OCO-2 selector uses this coordinatewise empirical operation with $M = 6$, $q = 40$ and $m_v = 2000$. Its candidate checkpoints and kernel heads were themselves selected on the same validation block, and that block has a different distribution from the test block. The independent-validation bound therefore does not apply to this fitted selector.

S2.1 A finite-design bound for truncating an input map

The next result separates the error from truncating the input map from the kernel error on the resulting lower-dimensional design.

Proposition S2.5 (finite-design rank truncation)**.** *Let $\mathcal{U} \subset \{u \in \mathbb{R}^d : \|u\|_2 \le R\}$ and let $G(u) = g(Au)$, where $A \in \mathbb{R}^{d\times d}$ has singular values $\sigma_1 \ge \cdots \ge \sigma_d \ge 0$ and $g : \mathbb{R}^d \to \mathbb{R}^q$ is L-Lipschitz in Euclidean norm. For $1 \le r \le d$, write $A_r = U_r \Sigma_r V_r^\top$ for a truncated singular-value decomposition, put $\sigma_{d+1} = 0$, and define*

$$\varphi_r(u) = \Sigma_r V_r^\top u, \qquad h(z) = g(U_r z), \qquad G_r = h \circ \varphi_r, \qquad \varepsilon = LR\sigma_{r+1}.$$

Suppose the components of h belong to the RKHS $\mathcal{H}_k$ of a positive definite kernel on $\mathbb{R}^r$, with $\|h\|_K^2 = \sum_{j=1}^q \|h_j\|_{\mathcal{H}_k}^2 < \infty$. For any design $X = (u_1, \dots, u_n)$ in $\mathcal{U}$, let $Z = (\varphi_r(u_1), \dots, \varphi_r(u_n))$, $K = k(Z, Z)$ and

$$c_\lambda(u) = (K + n\lambda I)^{-1} k(Z, \varphi_r(u)), \qquad \widehat{G}_\lambda(u) = c_\lambda(u)^\top G(X),$$

where $\lambda \ge 0$, and at $\lambda = 0$ the inverse is interpreted as a pseudoinverse when necessary. Let $\widetilde{P}_\lambda(u)$ be the factor in Theorem 6.9, evaluated in this feature space; equivalently,

$$\widetilde{P}_\lambda(u) = \left\| k(\varphi_r(u), \cdot) - \sum_{i=1}^n c_{\lambda,i}(u) k(\varphi_r(u_i), \cdot) \right\|_{\mathcal{H}_k}.$$

Then, for every $u \in \mathcal{U}$,

$$\|G(u) - \widehat{G}_\lambda(u)\|_2 \le \|h\|_K \, \widetilde{P}_\lambda(u) + \varepsilon\big(1 + \|c_\lambda(u)\|_1\big).$$

Proof. See Supplement S8.18.

The first term is the native-space bound on the projected design. The second includes both the pointwise approximation error from discarding directions and its propagation through the regression weights. For $\lambda > 0$ and bounded kernel diagonal, Lemma S1.7 also bounds the latter weights uniformly. Sampling rates and comparisons with the unprojected kernel require additional control of the power function and the target norm. The estimated sensitivity ranks in the OCO-2 experiments do not supply that control.

S3 Coverage of calibrated prediction balls

S3.1 Coverage under split-conformal assumptions

Split-conformal calibration rescales the positive function P_λ by a quantile of the normalized residuals. Write $\widehat{G}$ for the fitted predictor, $e(u) = G(u) - \widehat{G}(u)$, and let

$$s_i = \frac{\|e(\tilde{u}_i)\|_2}{P_\lambda(\tilde{u}_i)}, \qquad k = \lceil (1-\alpha)(m+1) \rceil, \qquad Q = \begin{cases} s_{(k)}, & k \le m, \\ +\infty, & k = m+1, \end{cases} \tag{S3.1}$$

where $s_{(k)}$ is the k-th order statistic of the m calibration scores and $0 < \alpha < 1$. When the required index exceeds the calibration sample size the prediction set is the whole output space.

Proposition S3.1 (finite-sample coverage). *Fix the predictor $\widehat{G}$, design X and positive scale function P_λ. Suppose the calibration and test inputs $\tilde{u}_1, \ldots, \tilde{u}_m, u^\star$ are exchangeable and independent of these fitted objects. Define Q by* (S3.1) *and*

$$C(u) = \{v : \|v - \widehat{G}(u)\|_2 \leq QP_\lambda(u)\}.$$

Then $\mathbb{P}(G(u^\star) \in C(u^\star)) \geq 1 - \alpha$. If the scores are almost surely distinct, their coverage is exactly $k/(m+1)$, and hence is at most $1 - \alpha + 1/(m+1)$. When $k = m+1$, the coverage is one, regardless of ties.

Proof. See Supplement S8.21.

Remark S3.2 (conditional reference law for realized coverage). *Suppose, more strongly, that conditional on the fixed fitted predictor and scale function, calibration and future evaluation scores are i.i.d. from a continuous distribution F. If $1 \leq k \leq m$, the coverage probability conditional on the calibration sample is $F(s_{(k)})$. The probability integral transform makes $F(s_i)$ i.i.d. uniform, so*

$$F(s_{(k)}) \sim \mathrm{Beta}(k, m+1-k).$$

This is the standard order-statistic law (Vovk, 2012). Conditional on the calibration sample, the count of covered cases in an independent evaluation set of size N is binomial with success probability $F(s_{(k)})$; integrating gives a Beta–binomial count. For $k = m+1$ the conditional coverage is instead identically one and the count is N; there is no Beta distribution with a zero second parameter. Exchangeability alone is sufficient for Proposition S3.1, but does not imply this stronger Beta–binomial law.

The proposition follows from the usual split-conformal rank argument (Vovk et al., 2005; Lei et al., 2018). The score is a residual norm divided by a positive scale, and the resulting set is a Euclidean ball in output space with radius $QP_\lambda(u)$. The statistic `mean_width` is the mean radius. The guarantee is marginal over a new input and concerns the output vector jointly; it does not give conditional coverage at each input or coordinatewise intervals. Constant-radius and other positive-scale scores satisfy the same proposition under the same independence assumptions, whether or not the scale ranks error well.

The experiments apply this calculation to a 1000/19000 partition of the public test block after fitting the models. The six-member coverage is 89.9±0.8% at the nominal 90% level, with P_λ rebuilt from each seed's correction kernel (`campaign/uq_conformal_plam.py`). Under the i.i.d.-continuous reference model, $m = 1000$, $\alpha = 0.1$ give $\mathrm{Beta}(901, 100)$ conditional coverage, and the central 95% reference interval for the fraction on $N = 19000$ cases is $[88.0\%, 91.8\%]$; at the 95% nominal level it is $[93.5\%, 96.3\%]$. All ten seeds fall inside these intervals at both levels. The seeds share test cases and are not ten independent draws from this law.

Under the proposition's sampling assumptions, coverage requires neither native-space membership nor a known residual norm, and remains valid when P_λ ranks errors poorly.

S4 Additional benchmarks and reflection symmetry

S4.1 Operator-learning comparisons

A common training protocol was applied to the other problems of the operator-learning suite that Batlle et al. (2024) assembled from de Hoop et al. (2022) and Lu et al. (2022): a Matérn

Table S4.1: Additional benchmark survey. Published reference mean relative L^2 test error on each problem (per-problem methods tuned by their authors), compared with the common training protocol on training splits and grids that differ from the baselines' in the cases noted in the text: single runs, except the two rows marked three seeds (mean $\pm$ standard deviation, exact solve on 19,000 pairs, output rank 400).

Problem	Map	Published reference	This work
Burgers	initial condition $\to$ solution	1.93% (FNO)	2.38%
Darcy flow	permeability $\to$ pressure	2.32% (POD-DeepONet)	2.01%
Advection I	smooth pulse $\to$ solution	≈ 0% (kernel)	—
Advection II	discontinuous pulse $\to$ solution	11.28% (linear kernel)	11.29%
Helmholtz	wavespeed $\to$ field	1.00% (kernel)	1.17% $\pm$ 0.02 (three seeds)
Navier–Stokes	vorticity $\to$ vorticity	0.12% (kernel)	0.28% $\pm$ 0.00 (three seeds)

kernel at the median length scale with the fit capped at 6000 points, a residual MLP mean, outputs reduced by PCA, and the stage selected on validation. Table S4.1 lists the results next to the reference values collected in that study. These comparisons use different tuning budgets and, in some cases, different data configurations. The published methods were tuned separately for each problem. The present results are single runs except for Helmholtz and Navier–Stokes, which use three seeds, exact kernel solves on 19,000 training pairs, and output rank 400. Rank 40 gives PCA reconstruction errors of 1.7% and 8.2% on these two problems; rank 400 reduces both below 0.01%, well below the prediction errors. Burgers and Darcy use 800 and 824 training pairs, respectively, slightly fewer than their baselines, and Darcy also uses a different grid. No result is reported for Advection I.

The kernel stage is selected on Helmholtz, Navier–Stokes, and Darcy. Its errors range from 1.2 times the tuned reference error on Helmholtz to 2.3 times that on Navier–Stokes. This protocol does not use the reference kernels' Cholesky preconditioning of the input. On Burgers, the corrected mean has an error of 2.38%, compared with 1.93% for the published Fourier neural operator. On Helmholtz the per-coordinate selection improves on the kernel alone (1.17% against 1.22%) by assigning a small subset of the 400 output coordinates to the corrected mean; on Navier–Stokes it selects the kernel almost everywhere.

S4.2 Empirical reflection symmetry

The continuous problem is invariant under $x_1 \mapsto 1 - x_1$: the domain and boundary partition are symmetric, and the input law has a symmetric covariance. On the grid, reflecting the load (S) should reflect the stress field along the first grid coordinate (T), i.e. $G(Su) = TG(u)$. For each of the first 200 samples we searched all 40000 loads for the nearest neighbor of the reflected load Su_i and compared the corresponding outputs. For the five best-matching pairs, the input mismatch $\|Su_i - u_j\|/\|u_j\|$ ranges over 0.27–0.32, while the output mismatch $\|Tv_i - v_j\|/\|v_j\|$ ranges over 0.085–0.14; reflecting along the second coordinate instead gives output mismatches of order one. The smaller mismatch along the first coordinate supports that choice of output reflection axis. The empirical mean and covariance of the training loads are invariant under reflection to within 1.1% and 1.5% respectively. These moment checks are consistent with, but do not establish, the exact input-law invariance assumed by Proposition S1.1. Reflection averaging at test time improves twenty-nine of the thirty runs of the two MLP variants and the refiner across ten seeds (Section 4). For the full-map average, Proposition S1.1 controls expected loss under its exact assumptions; a finite test sample can show an increase. For the ten refiner runs, identifying the supplied-field average with the full-map average requires the channel-equivariance condition in Supplement S1.1.

S5 Effective dimension from the Gram spectrum

S5.1 Spectral identities

The nugget λ, selected by cross-validation, determines the effective dimension through the spectrum of the Gram matrix.

Lemma S5.1 (effective dimension and the empirical Stieltjes transform)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$, let $\widehat{m}(z) = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i/n - z)^{-1}$ be the Stieltjes transform of the empirical spectral distribution of K/n. Then for every $\lambda > 0$ the effective dimension of ridge regression with nugget λ satisfies*

$$d_{\mathrm{eff}}(\lambda) \;:=\; \operatorname{tr}\bigl(K(K+n\lambda I)^{-1}\bigr) \;=\; n\bigl(1-\lambda\,\widehat{m}(-\lambda)\bigr).$$

In particular, if the empirical spectral distribution of K/n converges weakly to a limit law F as $n \to \infty$, then $d_{\mathrm{eff}}(\lambda)/n \to 1 - \lambda\, m_F(-\lambda)$ with m_F the Stieltjes transform of F.

Proof. See Supplement S8.23.

The identity holds for every positive semidefinite Gram matrix. For isotropic i.i.d. linear features, the limiting spectral law gives a closed form.

Corollary S5.2 (effective dimension under a Marčenko–Pastur limit)**.** *Let $K = XX^\top$, where $X \in \mathbb{R}^{n\times p}$ consists of the leading entries of an infinite array of i.i.d. real random variables with mean zero and finite variance $\sigma^2 > 0$. Let $p/n \to \gamma \in (0,1]$. For each fixed $\lambda > 0$,*

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} \;\longrightarrow\; \gamma\bigl(1-\lambda\, m(-\lambda)\bigr),$$

where

$$m(-\lambda) = \frac{\sqrt{\bigl(\lambda+\sigma^2(1-\gamma)\bigr)^2 + 4\gamma\sigma^2\lambda} - \bigl(\lambda+\sigma^2(1-\gamma)\bigr)}{2\gamma\sigma^2\lambda},$$

almost surely, where m is the Stieltjes transform of the Marčenko–Pastur law with ratio γ and scale σ^2.

Proof. See Supplement S8.22.

In one Gaussian realization with $n = 3000$, $p = 900$, and $\sigma^2 = 1.7$, the largest absolute difference between the limit formula and the empirical $d_{\mathrm{eff}}(\lambda)/n$ was 4.4×10^{-5} over the five values $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$.

The closed form is a reference for the stated i.i.d. linear-feature model. The learned neural features and nonlinear kernel matrices in the experiments lie outside those assumptions; the finite-matrix trace identity holds regardless. Supplement S6.3 reports its application to a 6000-row subsample of the structural-mechanics training design.

The effective dimension also determines the sum of the training-point posterior variances. Writing $A = K + n\lambda I$, their sum is

$$\sum_{i=1}^{n} P_\lambda(u_i)^2 = \mathrm{tr}(K) - \mathrm{tr}(KA^{-1}K) = \mathrm{tr}\big(n\lambda\, KA^{-1}\big) = n\lambda\, d_{\mathrm{eff}}(\lambda),$$

using $KA^{-1}K = K - n\lambda\, KA^{-1}$. This identity concerns the training design. The trace alone does not determine the power function at other inputs.

S6 Additional structural-mechanics results

S6.1 Ablation

Table S6.1 gives the contribution of each pipeline component. Kernel ridge on loads and the reflection-averaged neural mean have test errors 5.197% and 4.836%, respectively. With a single mean, stacking leaves the prediction unchanged; combining several means and correcting their residuals both reduce error. The feature-space and second input-space corrections did not improve validation error beyond the first correction with a single mean and so were not selected.

Reflection test-time averaging lowers the metric-trained MLP's test error by 0.16 points. The reduction is consistent with Proposition S1.1, whose exact symmetry assumptions are stronger than the empirical agreement measured in Supplement S4.2. The effect is smaller on the normalized-MSE network (0.02 points). Across the ten refiner seeds, the implemented supplied-field average lowers test error by 0.0422 ± 0.0019 percentage points (mean and sample standard deviation), with an improvement at every seed. Random reflection augmentation during training does not make the averaging effect vanish. The refiner's supplied-field operation and the full-map average are distinguished in Supplement S1.1. The kernel-flow regularizer did not improve the low-data MLP's validation error. At high data, mean test errors without reflection averaging are 4.99% at weight 0.3 and 4.99% without the regularizer, each over ten seeds. Weight 1.0 gives 5.05% over five seeds under the same scoring convention.

Replacing the single tuned Matérn correction by a sum at several bandwidths leaves validation error unchanged to two decimals under the stated search and tuning budget. In the five-member experiment, nine seeds select the largest nugget on the grid, $\lambda = 10^{-3}$, and one selects 10^{-5}. The length-scale multiplier is 2 at five seeds, 1 at four and 4 at one. The four-member ablation selects multiplier 2 at eight seeds.

The kernel prediction supplies the refiner's conditioning field. Its residual is less correlated with the neural members' residuals on the measured splits, which motivates this input. Section 4.5 reports the ensemble comparisons and their second-moment diagnostics.

S6.2 Spatial structure of the residual error

Ranked by relative error on the validation split, the twenty hardest cases have error 10.24% for the reference member against a median of 4.44%, a factor of 2.31. On those same cases the five-member ensemble reaches 10.18% and the mean over individual members is 10.28%. The ensemble improves this tail only slightly. Because these cases were selected by the reference member's errors, the comparison describes that selected subset.

Table S6.1: Component contributions in the high-data pipeline. Mean relative L^2 test error under the 20000-sample protocol; "+" rows are cumulative. All rows are mean $\pm$ standard deviation over ten seeds; the last two are the second experiment of Section 4.1.

Stage	Test error (%)
Kernel ridge on loads	5.197 ± 0.003
Reflection-averaged neural mean	4.836 ± 0.018
+ refiner and MSE variant, affine stack	4.628 ± 0.009
+ residual kernel correction	4.611 ± 0.005
+ FNO member, affine stack	4.595 ± 0.008
+ residual kernel correction	4.572 ± 0.010
+ complete FNO schedule and UNet member, affine stack	4.567 ± 0.004
+ residual kernel correction	4.546 ± 0.003

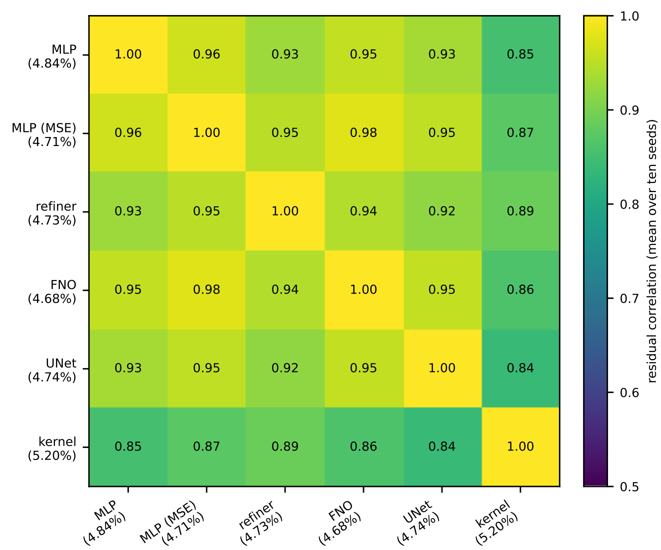

Figure S6.1: Centered correlation of per-sample normalized residuals on the validation split, six members, averaged over the ten seeds of the second experiment. Across seeds and pairs the entries range between 0.82 and 0.98. The mean over pairs is 0.919 ± 0.005 across seeds; the kernel predictor has correlation 0.86–0.88 with the normalized-MSE network at every seed. These centered correlations differ from the uncentered alignments summarized in Section 4.5.

The root-mean-square pointwise error over validation samples is 8.99 on the perimeter of the 41×41 grid and 5.64 in the interior, a ratio of 1.59; both vary across seeds by less than one percent of their values. Averaging lowers them to 8.87 and 5.59. The concentration near the loaded edge and supports remains visible in Figure 1. The shared spatial structure does not distinguish approximation bias from numerical artifacts in the target fields.

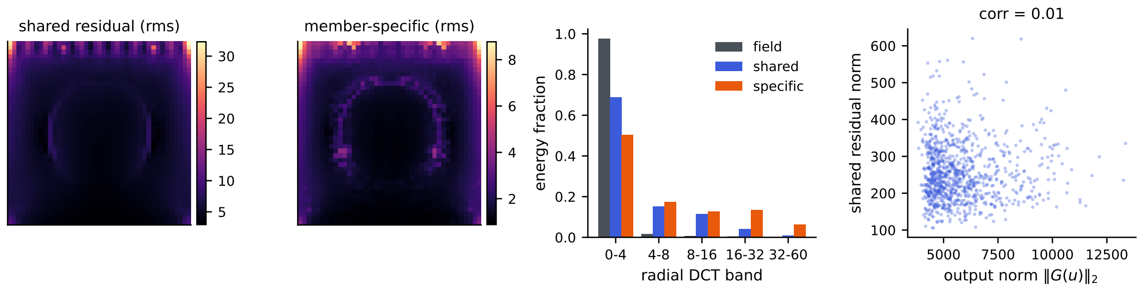

Figure S6.2: Spatial and spectral structure of the shared residual c (across-member mean of the residual fields), at one seed. Left to right: rms of c over the grid; rms of the member-specific remainders; radial DCT band energies of the stress fields, of c, and of the member-specific remainders; and the norm of c against the output norm, whose correlation is 0.01.

S6.3 Spectra and effective dimension

A separate diagnostic uses 6000 rows sampled without replacement from the 19000 training rows. The load coordinates are standardized using the full training design. The Matérn-5/2 length scale is fixed at four times the median distance on a 2000-row training subsample, and the marked nugget is 10^{-5}. These fixed diagnostic parameters need not match those selected by each deployed correction. The split and sampling generators both use seed 0.

The effective dimension is $d_{\text{eff}}(10^{-5}) = 103.11885$, where each eigenvalue μ_i contributes $\mu_i/(\mu_i + 6000\lambda)$. The Cholesky calculation of $6000 - 6000\lambda\,\mathrm{tr}\left((K_S + 6000\lambda I)^{-1}\right)$ agrees with the eigenvalue sum to 1.7×10^{-11}. This calculation describes the subsample smoother rather than the full 19000-row fit. Low effective dimension does not reduce the cubic work of the dense factorization used here.

S6.4 Computation and memory

The structural-mechanics experiments use cloud CPU instances and dense, exact linear solves for the kernel stages. At the experimental dimensions, an eight-thread timing calculation measured 27 seconds to build a 19000×19000 Gram matrix in blocks, 22 seconds for its Cholesky factorization, and 11 seconds to solve for 1681 output coordinates. The roughly one-minute total excludes hyperparameter search, network training and prediction. Values in this timing experiment are synthetic; the dimensions and operations match the experiment. The factorization performs $n^3/3 \approx 2.3 \times 10^{12}$ floating-point operations. No GPU, inducing points, preconditioner or stochastic linear solver is used in the correction.

Gram-matrix construction is the principal memory constraint at this size. Forming a full squared-distance matrix and its elementwise transforms allocates several 2.9 GB temporaries at once; in the experiments this construction required more than the available memory on a 31 GB instance. The block-filled timing diagnostic has a measured peak of 6.0 GiB, including the factorization and solves, compared with 19.2 GiB after the full-matrix construction. The reported pipeline uses full-size temporaries; the lower peak belongs to the separate block-filled implementation. The neural means have a few million parameters and include both metric-loss and normalized-MSE training. Splits are specified by permutation seeds and inputs are the 41-dimensional loads described in Section 2.

S6.5 Uncertainty quantification

Theorem 6.9 gives a conditional bound for a residual function in the kernel native space with consistent training labels. The deployed correction uses mixed training predictions: the kernel channel is constructed with foldwise fits whose target centering uses the pooled training labels, while neural predictions on these rows are in-sample. These labels need not equal $G(X) - m_{\text{full}}(X)$ for the deployed mean. The theorem applies to the deployed error under its stated consistency and native-space assumptions and, where applicable, the mismatch term.

Table S6.2 reports a computable diagnostic: the squared RKHS norms of ridge functions fitted to the two label arrays. With the selected correction kernel at seed 0, the residual-label fit has a norm about $4900\times$ smaller than the raw-stress fit and about $8.8\times$ smaller per unit label energy. These ratios depend on the nugget and the scale. The mixed residual labels prevent identifying this fitted norm with a lower bound for the native-space norm of the deployed residual $G - m_{\text{full}}$.

The design factor P_λ correlates weakly with the deployed surrogate's absolute error (Pearson 0.055) and negatively with relative error. Its correlation with the output norm is 0.65 (Figure S6.3), so its association with error depends on normalization by output amplitude. Ensemble disagreement has correlation 0.074 ± 0.006 with absolute error across ten seeds and -0.25 ± 0.04 with relative error. The corresponding four-member absolute error correlation is 0.083. Thus neither signal ranks the remaining error well. A spread statistic is insensitive to a common additive error shared by all members, which is consistent with the residual alignment observed here.

Three bands are calibrated on 1000 cases selected from the test block and scored on the other 19000: the scores are $\|e\|_2/P_\lambda$, $\|e\|_2$ (constant radius), and $\|e\|_2$ divided by ensemble disagreement. Each seed's P_λ uses its own selected correction kernel and training design. The measured coverages are split diagnostics on the public test block. The marginal split-conformal theorem applies when the predictor and score are fixed independently of exchangeable calibration and future observations; the exact Beta and Beta–binomial descriptions additionally require conditionally i.i.d. continuous scores (or an appropriate randomized rank construction). A random partition of a fixed benchmark pool alone does not supply these sampling assumptions.

For the six-member pipeline the P_λ band has empirical coverage $89.9 \pm 0.8\%$ at the nominal 90% level and $95.0 \pm 0.6\%$ at 95%. Under the i.i.d.-continuous reference model, $m = 1000$ gives latent coverage $\text{Beta}(901, 100)$ at the 90% level; with 19000 independent evaluation scores its central 95% Beta–binomial reference interval is $[88.0\%, 91.8\%]$. At nominal 95% coverage, the corresponding central 95% reference interval is $[93.5\%, 96.3\%]$. All ten six-member coverages fall inside these intervals at each level.

The constant-radius band covers $90.4 \pm 0.8\%$ and $95.4 \pm 0.5\%$, inside the reference intervals at ten of ten seeds at both levels; the P_λ band is on average 1.11 times as wide. The disagreement-scaled band covers $90.1 \pm 0.4\%$ and $95.0 \pm 0.8\%$, with one seed on the upper edge of the 95% reference interval. The five-member pipeline gives $90.1 \pm 0.9\%$ and

Table S6.2: Fitted-norm diagnostic of the residual correction at seed 0 of the complete-schedule experiment: the same validated correction kernel (scale multiplier 2, nugget 10^{-3}, median length scale) ridge-fitted to the raw stress fields and to the six-member stack residuals on the training split. $\|\widehat{r}\|_K^2 = \mathrm{tr}(\alpha^\top K \alpha)$ is the squared RKHS norm of the fitted function; the last column divides it by the energy of the fitted label array. The residual labels mix foldwise kernel predictions and in-sample neural predictions, so these numbers are not identified as lower bounds for the native-space norm of the deployed residual. The ratios depend on the selected nugget and scale. Much of the raw ratio reflects output amplitude: per unit energy the residual's fitted norm is 8.8 times smaller at this seed, not 4900; at seeds 1 and 2, whose corrections chose scale multipliers 1 and 2, the raw ratios are 1800 and 4300 and the normalized ones 3.2 and 7.6.

Kernel regresses	$\|\widehat{r}\|_K^2$	energy $\|T\|_F^2$	$\|\widehat{r}\|_K^2/\|T\|_F^2$
Raw stress fields ($m = 0$)	7.12×10^8	6.77×10^{11}	1.05×10^{-3}
Ensemble residuals	1.45×10^5	1.21×10^9	1.20×10^{-4}
Ratio	$4900\times$	$560\times$	$8.8\times$

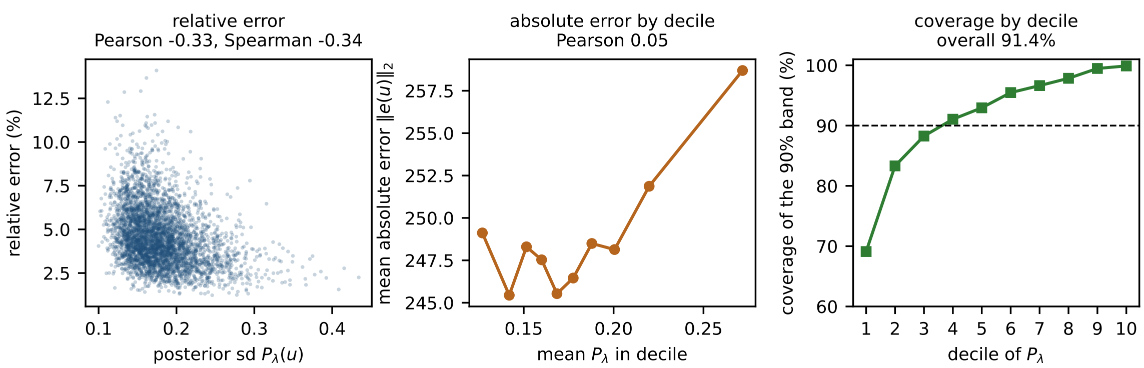

Figure S6.3: Uncertainty measures for the six-member correction at seed 0. Calibration uses 1000 test-block cases and evaluation uses the other 19000, as in Section 4.7. Left: the Gaussian process posterior standard deviation P_λ against relative error; the association is negative (Pearson -0.33) because large-amplitude outputs have small relative error. Middle: decile calibration of P_λ against absolute error $\|e(u)\|_2$; the association is weak (Pearson 0.05). Right: coverage of the P_λ-scaled 90% band within each decile of P_λ: 91.4% overall, but from 69% in the lowest decile to 99.9% in the highest, so coverage varies strongly with the design factor despite an overall value close to the nominal level. Across the ten seeds the same band has coverage $89.9 \pm 0.8\%$ at the nominal 90% level, inside the i.i.d.-continuous Beta–binomial reference interval at ten seeds of ten; the constant-radius band is the narrower of the two (Section 4.7).

$95.1 \pm 0.8\%$ for the P_λ band (ten and nine seeds inside), and $90.6 \pm 0.4\%$ and $95.4 \pm 0.6\%$ for the constant-radius band (ten inside at both levels). The constant-radius band is narrower, and the kernel design factor provides no measured efficiency gain.

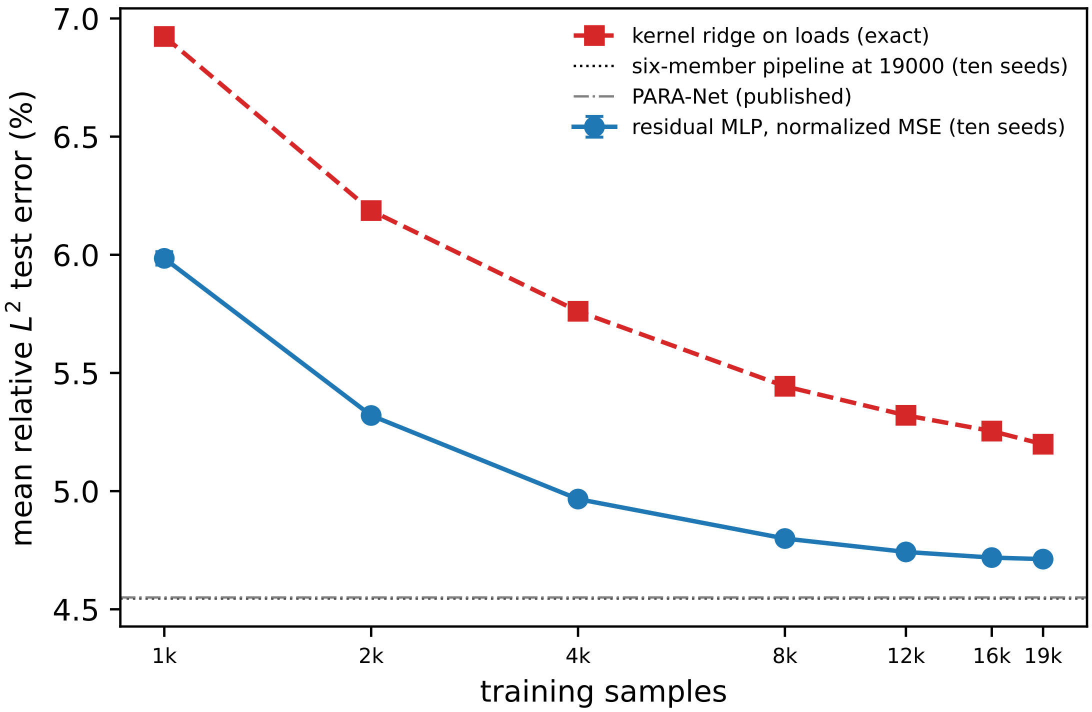

Figure S6.4: Structural-mechanics learning curves. The first $N + 1000$ rows of the training pool are used, with the final 1000 reserved for validation and the first N for fitting. The validation split is fixed across seeds at each N and changes with N; evaluation uses the full test block. The curves give mean relative test error for the normalized-MSE network (mean and standard deviation over ten seeds) and deterministic Matérn kernel ridge on loads, with both axes logarithmic. The horizontal lines mark the six-member pipeline at 19000 samples and the quoted PARA-Net result. Error bars are smaller than the markers above 4000.

S6.6 Seed variation and test-sample uncertainty

The seed standard deviations describe variation from initialization and selection on a common fixed test block. They do not include the sampling uncertainty of a future evaluation set. If cases were independent and representative of a target population, the reported per-case standard deviation 0.0169 would give a mean-error standard error of 0.012 percentage points over 20000 cases.

The published baselines do not provide per-case predictions or comparable training replicates. Their differences from our scores therefore do not support a paired significance test or an equivalence conclusion. Combining two copies of our sampling term in quadrature gives approximately 0.017 points, but this imputes an unknown baseline variance and covariance structure. It is a sensitivity calculation. The 0.098-point difference from PCA-Net and 0.022-point difference from PARA-Net for the five-member pipeline are benchmark-score differences whose statistical uncertainty cannot be determined from the available baseline summaries.

Within the two experiments, per-case paired differences isolate changes between fitted predictors on the same cases. The full pipeline reduces error by 0.140 percentage points, and removing the FNO increases it by 0.039 ± 0.011 points across ten seeds, with the latter direction holding at all ten.

The validation second-moment simplex optimum distributes weight among the five members: refiner 0.31 ± 0.04, FNO 0.34 ± 0.06, normalized-MSE network 0.24 ± 0.08, kernel 0.06 ± 0.03, and metric-trained MLP 0.06 ± 0.04. The kernel has nonzero weight at all ten seeds and the metric-trained MLP at nine. For the six-member experiment the corresponding shares are UNet 0.32 ± 0.06, FNO 0.30 ± 0.07, refiner 0.27 ± 0.04, kernel 0.06 ± 0.02, normalized-MSE network 0.03 ± 0.04, and metric-trained MLP 0.02 ± 0.02. The FNO's large share despite its single-member ranking illustrates the role of off-diagonal residual second moments.

Proposition 6.1(i) uses the uncentered alignment $S_{12}/\sqrt{S_{11}S_{22}}$. The descriptive centered correlations in Figure S6.1 need not equal this alignment. Exact convex-mixture calculations use the uncentered second-moment matrix, rather than the centered-correlation summaries.

The completed FNO schedule reduces its validation RMS error from 5.047% to 4.991%, while its descriptive correlation with the normalized-MSE network increases from 0.95–0.97 to 0.974–0.980. The five-member validation quadratic optimum is 4.940% before and 4.940% after this change. These measured changes are consistent with an accuracy improvement being offset by greater residual alignment. Proposition 6.7 gives a local sensitivity calculation for fixed positive RMS errors and uncentered alignment; the centered-correlation summaries do not determine that sensitivity.

For an exactly equicorrelated equal-error family, Proposition 6.1(ii) gives the limiting RMS value $\bar{e}\sqrt{\bar{\varrho}}$. Substituting the measured mean RMS error gives the heuristic value $4.922 \pm 0.056\%$ for five members using the mean centered correlation, and $4.908 \pm 0.057\%$ for six members using the mean uncentered alignment. The empirical pools are heterogeneous, so these substitutions are heuristic diagnostics. The one-sided entrywise bounds below and the sixty-member bound use the actual uncentered second-moment matrices instead.

All these bounds concern RMS error $\mathcal{E}_2$ for global convex weights. Jensen's inequality gives $\mathcal{E}_1 \leq \mathcal{E}_2$ for the same predictor, so an RMS lower bound does not transfer to mean error. The deployed pipeline's per-pixel affine weights and kernel correction also place it outside the global convex class. Among global sum-to-one weights the signed and convex optima differ by at most 0.00052 percentage points across the fitted pools, but they do not always coincide: the signed minimizer has a negative coordinate at seven six-member seeds and one five-member seed.

The selected input-space residual correction improves the five-member stack by 0.024 points at $n = 19000$. Figure S6.4 reports the observed response to sample size: the normalized-MSE network goes from $5.985 \pm 0.028\%$ at $N = 1000$ to $4.799 \pm 0.003\%$ at 8000 and $4.712 \pm 0.004\%$ at 19000; the kernel goes from 6.924% to 5.444% and 5.198%. Over the measured range above 8000, fitted log-log slopes are -0.022 ± 0.001 for the network and -0.052 for the kernel. The final increase in sample size reduces error by 0.006 ± 0.004 percentage points, with a reduction at nine seeds. The last two increases together reduce error by 0.030 ± 0.010 percentage points, with a reduction at all ten seeds. These are finite-range fits. A three-parameter fit $e_\infty + cN^{-\alpha}$ gives different extrapolated asymptotes, 4.64% and 4.88%, so it does not locate a common floor. Shared approximation bias and target discretization artifacts remain unresolved explanations for the residual plateau.

On the five-member test set, Corollary 6.4 gives a normalized bracket $[0.949 \pm 0.005, 1.039 \pm 0.004]$ and fitted-global-stack quadratic risk 0.980 ± 0.003. That risk uses the deployed global weights. The six-member validation-matrix calculation reports a bracket $[0.933 \pm 0.007, 1.032 \pm 0.004]$ and normalized simplex optimum 0.972 ± 0.005. The corollary upper-bounds the minimum by the uniform-weight risk; an arbitrary fitted ensemble need not lie below that upper bound, although the five-member weights do.

From the six-member validation matrices, the one-sided RMS floor $\sqrt{\min_{i \neq j} S_{ij}}$ is $4.820 \pm 0.062\%$, the stronger finite-six-member lower bound is $4.849 \pm 0.060\%$, and the quadratic optimum is $4.920 \pm 0.057\%$. The dispersion factors 0.938 ± 0.003 and 0.938 ± 0.003 compare $\mathcal{E}_1$ and $\mathcal{E}_2$ on the same validation data used to compute the weights. They measure the difference between the two metrics on those validation data. For six members the ratio of test mean error to validation RMS error is 0.9357 ± 0.0110; that ratio additionally includes transfer between samples and is not a pure dispersion factor.

The sixty-predictor pool. The analysis of Section 4.6 uses the test predictions of all sixty members in the second experiment. A fixed permutation divides the test block into 1000 fitting and 19000 evaluation cases for this pooling exercise. Convex weights fitted on the first subset are evaluated on the second; the hindsight optimum is explicitly optimized on the evaluation matrix. Equal-weight seed curves average over all subsets for RMS error; mean errors use all subsets where there are at most 45, and 40 random subsets otherwise.

Using the evaluation matrix, the smallest diagonal is 0.0024512070 and the within- and between-architecture lower second moments divided by that diagonal are 0.9807963 and 0.9377968. Theorem 6.6 then gives 4.814%, below the hindsight RMS optimum 4.876%. This empirical lower bound applies to global convex combinations of these sixty predictors on the evaluation cases. An extension to more seeds would require the same entrywise lower bounds to hold for those additional members.

Within the existing ten-seed blocks, square roots of the smallest within-architecture second moments are approximately 4.93% for the FNO, 4.96% for the normalized-MSE network, 4.97% for the refiner, 4.94% for the UNet, 4.90% for the metric-loss MLP and 5.46% for the kernel. The corresponding ten-seed equal-weight errors lie close to these empirical references. Replacing lower moments by mean moments in the block formula returns the uniform sixty-member mixture's value exactly, as an algebraic identity.

At the pipeline's fixed ridge, per-pixel affine fitting on all sixty members has evaluation error 4.682%, versus 4.556% for the six ten-seed means and 4.567% for six members at seed 0. The leave-one-out ridge comparisons in Section 4.6 show that this difference depends on regularization. Removing the UNet seeds changes the uniform-weight mean error from 4.611% to 4.630%.

For the first experiment the test-error seed standard deviation is 0.005 points at uniform weights, 0.006 for fitted global weights, 0.008 for the affine stack and 0.010 after correction.

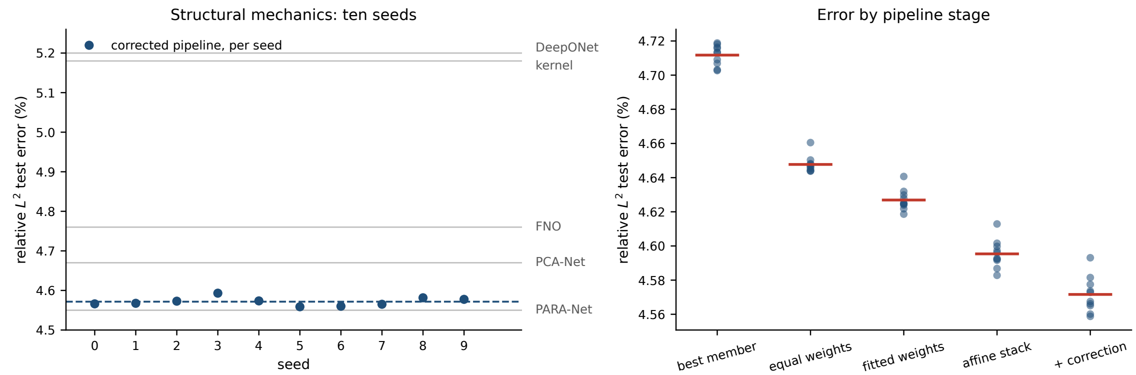

Figure S6.5: Left: the corrected pipeline's test error at each of the ten seeds (points), their mean (dashed), and the five published benchmark errors on the same axis (grey). These show the variability of our pipeline; comparable seed distributions are unavailable for the published methods. Right: test error at successive pipeline stages for each seed, with the stage means marked. Fitting per-pixel weights lowers the mean; seed spreads at each stage are reported in the text.

S7 OCO-2 input metrics and data scaling

S7.1 Input rescaling and prediction error

Both raw-input kernels are fitted at five nested training sizes over a sixteenfold range, at ten seeds on each band. Each sensitivity map is estimated from the training rows available at its own sample size using the standardized-input least-squares matrix A in Supplement S9. Rank summaries use the separate matrix $B = X_{\mathrm{tr}}^{\dagger} Z$ in the released input coordinates on all 18000 fitting rows. The singular-value counts retaining 99% of B's squared energy are 12/20, 14/24 and 13/24; the corresponding participation ratios $(\sum_j \sigma_j)^2 / \sum_j \sigma_j^2$ are 11.99, 13.63 and 13.33. These describe the fitted linear maps in those coordinates.

Table S7.1 reports finite-range log–log slopes. On O2 the paired slope difference is small relative to its seed variation. The adapted kernel's fitted exponent is larger on the two CO_2 bands, by 0.011 and 0.052. At $n = 18000$ the isotropic-to-adapted error ratios are 1.353 ± 0.008, 1.118 ± 0.002 and 1.514 ± 0.114. Thus the tested rescaling improves prediction at that training size but leaves much of the error difference between raw-input and feature kernels. These finite-range slopes are distinct from the rank-truncation bound of Proposition S2.5, which requires Lipschitz, domain-radius, and native-space assumptions beyond the estimated ranks.

At $n = 18000$, the sweep and Table 6 use the same tuning routine and standardized inputs and give identical raw-kernel results and selected hyperparameters at every seed. Training duration affects the comparison: a separate single-seed 750-epoch O2 run reaches 3.82% (flat network) and 3.71% (feature head), with the combination at 3.71% reduced and 0.0174% radiance. The 250-epoch comparison is therefore sensitive to training duration and does not establish convergence.

The training-loss comparison further distinguishes coordinate weighting from per-case weighting. The second network in Table 6 is trained on a diagonally weighted loss over

Table S7.1: The raw-input kernel refit at nested training sizes, ten seeds per band. "Rank" counts singular values of the fitting-set matrix $B = X_{\mathrm{tr}}^{\dagger} Z$ holding 99% of the squared energy, out of the input dimension. The exponent difference is paired within seed; its standard deviation is over seeds. The last column is the isotropic-to-adapted error ratio at $n = 18000$. The slopes are descriptive fits over the finite sizes tested.

Band	Rank	−slope iso	−slope adapted	difference	ratio at n=18000
O2	12/20	0.221 ± 0.007	0.215 ± 0.017	$+0.006 \pm 0.015$	1.353 ± 0.008
weak CO_2	14/24	0.086 ± 0.005	0.097 ± 0.006	-0.011 ± 0.009	1.118 ± 0.002
strong CO_2	13/24	0.203 ± 0.029	0.255 ± 0.020	-0.052 ± 0.021	1.514 ± 0.114

the reduced coefficients, $\|s_z \odot (\hat{z} - z)\| / \|s_z \odot z\|$. That is a surrogate for the radiance-relative error: the reported radiance metric maps predictions through the stored PCA reconstruction and divides by the reconstructed-radiance norm, and the surrogate omits the reconstruction. The third network is trained in the exact radiance metric, reconstruction and denominator included. Its radiance error is higher than that of the surrogate-trained network: $0.0571 \pm 0.0057\%$ against $0.0442 \pm 0.0044\%$ on O2 and $0.093 \pm 0.008\%$ against $0.075 \pm 0.003\%$ on SCO2, the surrogate having lower error at all ten seeds on both bands. The ranking is reversed on WCO2 at eight of ten seeds ($0.064 \pm 0.015\%$ against $0.072 \pm 0.007\%$). The kernel heads built on their features repeat the pattern. Tripling the training budget narrows the O2 gap (0.0212% against 0.0201% at 750 epochs) without reversing it, but these finite-budget comparisons do not identify why one training loss performs better. By the reconstruction identity, the two losses share the same numerator and differ in their target-dependent denominator, which reweights cases. Both concentrate on the leading coefficients, while their optimization and generalization behavior can differ. The coordinate profile is consistent with this trade-off: the kernel-flow emulator predicts the leading coefficient to 0.0012 root mean square against the flat network's 0.0054, whereas the flat network is three to four times more accurate on the trailing thirty coordinates. The relative errors depend on both the band and training budget.

The flat-network seeds permit a second-moment analysis similar to that on structural mechanics. For each band, the matrix S contains the test-set second moments of the normalized residuals of the ten flat-network seeds. Mixture risks are measured in squared relative error, as in Proposition 6.1.

At uniform weights, $w^{\top} S w$ and the directly measured risk of the averaged prediction agree to 7×10^{-18}, 3×10^{-17} and 1×10^{-17} on the three bands.

Over the 45 seed pairs per band, the residual alignments are 0.787 ± 0.010 (O2), 0.971 ± 0.001 (WCO2) and 0.961 ± 0.001 (SCO2). An equicorrelation summary gives ten-member RMS relative errors of 4.419%, 17.997% and 9.267%, matching the recorded uniform-mixture risks at the displayed precision. The corresponding infinite-family values, 4.360%, 17.970% and 9.248%, are extrapolations under the equicorrelation model. They are not lower bounds for untrained seeds or for the architecture class. The exact finite-pool statement is the quadratic identity applied to the measured matrix S.

Comparisons with another readout must use the same metric, since S determines a squared-relative-error quantity and Table 6 reports the mean relative error $\mathbb{E}\,\ell$. Measured in

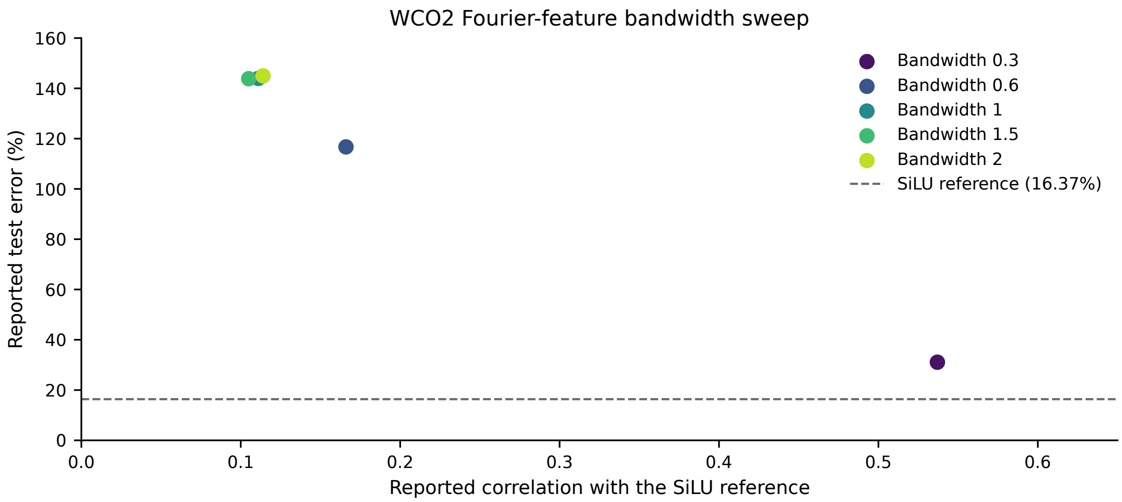

Figure S7.1: Exploratory WCO2 Fourier-feature sweep. The legend identifies bandwidths; the dashed line is the SiLU error of 16.37%. Three bandwidths nearly coincide in the upper left. Errors and correlations are rounded summaries of the exploratory measurements. The admission calculations in Section 5.2 use exact uncentered second moments.

RMS relative error, the Matérn head on learned features reaches 4.575%, 18.11% and 9.326% — better than an individual seeded network (4.915%, 18.238%, 9.433%) and worse than that network's ten-seed ensemble. The ranking is the same in the mean metric, where the seed ensemble reaches 3.916% against the head's 4.111% on O2. The ten-network ensemble has lower error than the single feature-kernel head in both metrics, but the predictors have different computational costs.

Lower residual alignment alone does not guarantee a useful mixture: the weaker member's RMS error also enters Proposition 6.1(i). An exploratory WCO2 Fourier-feature bandwidth sweep gives error near 31% with correlation 0.54, and errors above 140% with correlations near 0.11 (Figure S7.1). The plotted errors and correlations are rounded summaries; the admission-condition analysis in Section 5.2 uses the exact uncentered second moments of the eight-model pool. Adding the exploratory architecture candidates to coordinate selection changed the mean error by less than 0.01 percentage points on each band.

For the raw-input kernel against the flat network, the validation summary has $\varrho = 0.156$ and $e_n/e_k = 0.157$, with RMS relative errors 8.41% and 53.5%. The sign of the criterion changes across seeds, and the optimal mixture gains only a few ten-thousandths of a percentage point. This explains the pair's small contribution.

The combination chooses each coordinate from one of the three networks and three feature-kernel heads using coordinatewise validation RMSE. By Proposition S2.2, this minimizes an auxiliary separable squared loss over that candidate product. Fixed sample denominators retain separability for squared relative losses, with different sample weights for the reduced and radiance criteria. The reported means of relative norms instead couple coordinates through the square root. Consequently this selector is not guaranteed to optimize either reported metric, or both simultaneously. The same selected coordinates produce both reported columns; the baseline emulator's predictions are not candidates. Its seed-averaged errors are lower than those of the rescored kernel-flow emulator in both criteria on all three bands: O2 reaches $4.12 \pm 0.05\%$ against 16.89% reduced and $0.0294 \pm 0.0020\%$ against 0.0448% radiance, SCO2 $8.08 \pm 0.01\%$ against 16.14% and $0.0584 \pm 0.0041\%$ against 0.1147%, and WCO2 $16.28 \pm 0.06\%$ against 24.06% and $0.0507 \pm 0.0052\%$ against 0.0599%. Counted by seed, the combination has lower error than the emulator in both metrics at all ten seeds on O2 and SCO2 and at nine of ten on WCO2. At the remaining WCO2 seed, its radiance error is higher by 0.0015 percentage points.

On O2 the combination's reduced mean is slightly worse than that of the flat-feature head (4.11510% versus 4.11080%), while its radiance mean improves from 0.07927% to 0.02940%. The selector therefore improves radiance accuracy at a small cost in reduced-coordinate accuracy. On WCO2, one seed has a radiance error 0.0015 percentage points higher than the source emulator. The paired-bootstrap interval for that seed is $[-0.0012, 0.0042]$ percentage points, while the other nine intervals are negative. These are descriptive intervals on the public test cases.

S7.2 Learning curves and training budgets

The OCO-2 learning curve decreases more strongly with training size than the structural-mechanics curve over the ranges examined. The left panel of Figure S7.2 shows a one-seed O2 experiment with a disjoint held-out set. A finite-range log–log fit gives slope -0.68, with error decreasing from about 74% at a few hundred pairs to 4.3% at $n = 17700$. The main ten-seed combination at 18000 training pairs has mean 4.12% at its 250-epoch budget; the 3.71% result above belongs to a separate single-seed 750-epoch experiment. These are different protocols and should not be combined into one scaling fit. The raw-input kernel improves more slowly over these sizes. The measured curves do not identify the limiting source of error or an asymptotic convergence rate.

The ClimSim experiments cover a larger data range. ClimSim (Yu et al., 2023) is a climate emulation dataset of about ten million samples that maps a 124-dimensional atmospheric state to 128 physics tendencies, with published neural baselines. The runs compare two model families at one, two and three and a half million training samples. Kernel hyperparameters are selected on a disjoint 2000-row validation block drawn from the training pool. Each neural run retains the final checkpoint of its 60-epoch schedule. Evaluation uses separate data; the right panel of Figure S7.2 plots those points. Each experiment seed draws its own 20000-row test block, so the spreads below include that variation.

For each output coordinate j, let $v_j = N^{-1} \sum_{i=1}^{N} (y_{ij} - \bar{y}_j)^2 + 10^{-12}$ be its variance on that seed's test block, and define $J = \{j : v_j > 10^{-6} \max_k v_k\}$. The reported score is the unweighted mean of the coordinatewise coefficients of determination,

$$R^2 = \frac{1}{|J|} \sum_{j \in J} \left(1 - \frac{N^{-1} \sum_{i=1}^{N} (\hat{y}_{ij} - y_{ij})^2}{v_j} \right).$$

The same active-coordinate set J is used for both predictors at each seed.

The neural mean has higher R^2 than the kernel and improves with training size: 0.544 ± 0.005 at one million (five seeds), 0.563 ± 0.004 at two million and 0.575 ± 0.004 at three and a half million (three seeds each). A published multilayer-perceptron value near 0.6

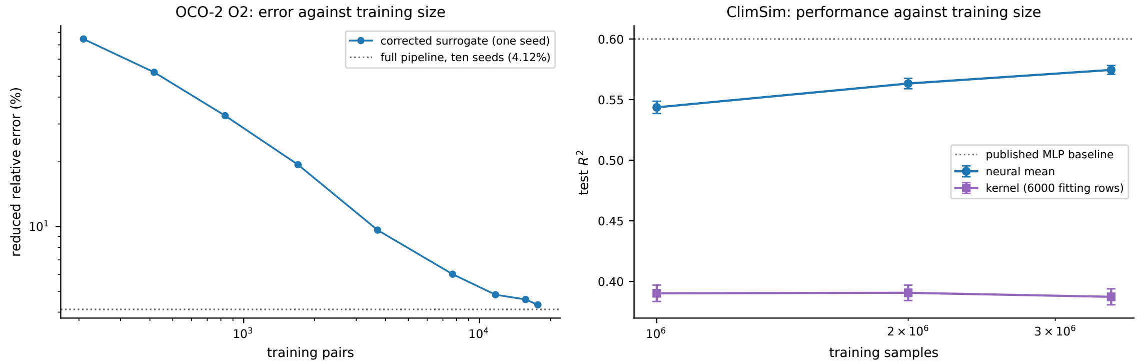

Figure S7.2: Error against training-set size. Left: the OCO-2 O2 task, reduced error of the corrected surrogate at one seed, from the data-scaling run; a finite-range log–log fit has slope -0.68, and the error is 4.3% at $n = 17700$, against the 4.12% the ten-seed pipeline reaches on all 18000 pairs (dotted). Right: ClimSim, test R^2 over the active outputs defined in the text, at one, two and three and a half million training samples — five seeds at one million and three at each of the larger sizes, error bars over seeds. No crossover between the two families is observed over this range. Each kernel is fitted on 6000 rows, so its nearly constant error across nominal training sizes reflects that fixed fitting budget. The separate kernel-budget comparison is reported in the text. The neural mean has higher R^2; the published baseline (dotted) is shown for orientation; the protocols are not matched.

provides context; this comparison does not use a matched split and scoring protocol. The corresponding kernel values are 0.390 ± 0.007, 0.391 ± 0.006 and 0.387 ± 0.007 at the three sizes. Each kernel was fitted on a 6000-row subsample, so the near-constant error does not describe its response to increasing kernel training size. A separate kernel-budget comparison uses the same protocol (selection on the pool validation block, separate test evaluation, seed 0, the one-million slice): 0.389 at 6000 rows, 0.420 at 12000 and 0.432 at 24000 (0.446 at a second seed, with a separately drawn test block). The R^2 value increases by about 0.03 at the first budget doubling and 0.01 at the second, and at 24000 rows, the largest exact solve tested, its R^2 remains about 0.11 below the network at one million samples. The comparison remains unequal in training size and total computation: the neural mean uses up to millions of examples, whereas the exact kernel uses at most 24000. The network has lower error under these budgets; matched-compute comparisons and larger kernel fits remain untested. The reported range does not locate a crossover between model families.

S8 Proofs

S8.1 Proof of Proposition S1.1

By convexity of $\ell(\cdot, v)$,

$$\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \tfrac{1}{2}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\ell\big(T\widehat{G}(Su), G(u)\big).$$

Since T is an involution, the equivariance $G(Su) = TG(u)$ applied at u gives $TG(Su) = T^2G(u) = G(u)$, hence

$$\ell\big(T\widehat{G}(Su), G(u)\big) = \ell\big(T\widehat{G}(Su), TG(Su)\big) = \ell\big(\widehat{G}(Su), G(Su)\big),$$

using the T-invariance of the loss. Taking expectations and using the S-invariance of μ (so that $Su \sim \mu$ when $u \sim \mu$),

$$\mathbb{E}\,\ell\big(\widehat{G}_S(u), G(u)\big) \le \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(Su), G(Su)\big) = \mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big). \quad \square$$

S8.2 Proof of Proposition S1.2

Strict positive definiteness of the restricted kernel matrix makes I_C the (unique) interpolant of the pairs indexed by C, so $I_C(z_j) = v_j$ for $j \in C$ and the terms of (S1.2) indexed by C vanish, which is the first claim. For the second, write

$$e_2(\theta; C) \;=\; \sum_{i \in B \setminus C} g(C, i), \qquad g(C, i) := \|v_i - I_C(z_i)\|_2^2.$$

Conditionally on C, the sum has exactly $b/2$ terms, so $e_2(\theta; C) = \frac{b}{2}\,\mathbb{E}_{i \sim \mathrm{Unif}(B \setminus C)}[g(C, i)]$, and taking the expectation over C proves the identity. The gradient claim follows from the differentiability assumptions: for fixed (B, C), differentiable feature and kernel maps make $\theta \mapsto e_2(\theta; C)$ differentiable wherever the kernel matrix in I_C remains positive definite. Under the stated local integrable domination, the derivative passes under the expectation, so $\mathbb{E}_{B,C}[\nabla_\theta e_2] = \nabla_\theta\,\mathbb{E}_{B,C}[e_2]$. $\square$

S8.3 Proof of Proposition S1.3

(i) Since $\sum_m w_m = 1$, $F_w(u) - G(u) = \sum_m w_m\,(f_m(u) - G(u))$, and the triangle inequality gives $\|F_w(u) - G(u)\|_2 \le \sum_m w_m \|f_m(u) - G(u)\|_2$. Dividing by $\|G(u)\|_2$ yields the pointwise claim; taking expectations gives the risk inequality, and bounding each $\ell(f_m(u), G(u))$ by B gives $\ell(F_w(u), G(u)) \le B$.

(ii) For $M = 1$ there is no weight selection and the asserted inequality is immediate. Suppose $M \ge 2$ and write $\ell_u(w) = \ell(F_w(u), G(u))$. First, ℓ_u is Lipschitz on Δ for the ℓ^1 norm: for $w, w' \in \Delta$, the reverse triangle inequality and the argument of (i) give

$$|\ell_u(w) - \ell_u(w')| \;\le\; \frac{\big\|\sum_m (w_m - w'_m)(f_m(u) - G(u))\big\|_2}{\|G(u)\|_2} \;\le\; B\,\|w - w'\|_1.$$

Next, discretize the simplex. For $k \in \mathbb{N}$ let $\mathcal{G}_k = \{w \in \Delta : kw \in \mathbb{Z}^M\}$; its cardinality is the number of compositions of k into M nonnegative parts, $\binom{k+M-1}{M-1} \le (k+1)^{M-1}$. Given $w \in \Delta$, set $w'_i = \lfloor k w_i \rfloor / k$ for $i < M$ and $w'_M = 1 - \sum_{i<M} w'_i$; then $w' \in \mathcal{G}_k$ (the last coordinate is a multiple of $1/k$ and is $\ge w_M \ge 0$ because the first $M-1$ coordinates only decreased), and

$$\|w - w'\|_1 = \sum_{i<M} (w_i - w'_i) + \big(w'_M - w_M\big) = 2 \sum_{i<M} (w_i - w'_i) \;\le\; \frac{2(M-1)}{k}.$$

By (i) the per-sample loss lies in $[0, B]$, so for each fixed w' Hoeffding's inequality gives $\mathbb{P}\big(|\widehat{R}(w') - R(w')| > t\big) \le 2e^{-2mt^2/B^2}$, and a union bound over $\mathcal{G}_k$ shows that with probability at least $1-\delta$,

$$\sup_{w' \in \mathcal{G}_k} \big|\widehat{R}(w') - R(w')\big| \;\le\; B\sqrt{\frac{\log\big(2|\mathcal{G}_k|/\delta\big)}{2m}}.$$

On this event, for every $w \in \Delta$, approximating by its grid point and using the Lipschitz property for both R and $\widehat{R}$,

$$\big|\widehat{R}(w) - R(w)\big| \;\le\; B\sqrt{\frac{\log(2|\mathcal{G}_k|/\delta)}{2m}} + \frac{4B(M-1)}{k}.$$

If $w^\star$ minimizes R over Δ, the empirical minimizer satisfies $R(F_{\widehat{w}}) \le \widehat{R}(\widehat{w}) + \sup_\Delta |\widehat{R} - R| \le \widehat{R}(w^\star) + \sup_\Delta |\widehat{R} - R| \le R(F_{w^\star}) + 2\sup_\Delta |\widehat{R} - R|$. Choosing $k = m(M-1)$ and using $|\mathcal{G}_k| \le (m(M-1)+1)^{M-1}$ gives

$$R(F_{\widehat{w}}) \;\le\; \min_{w \in \Delta} R(F_w) + \frac{8B}{m} + B\sqrt{\frac{2\big((M-1)\log(m(M-1)+1) + \log(2/\delta)\big)}{m}},$$

which is the claim. □

S8.4 Proof of Lemma S1.4

For each k the validation scores $\ell(g_k(\tilde{u}_i), G(\tilde{u}_i))$, $i \le m'$, are i.i.d., take values in $[0, B]$, and have mean $R(g_k)$; the maps g_k are fixed relative to this sample, which is the only place the independence assumption enters. Hoeffding's inequality gives, for each k,

$$\Pr\Big(\big|\widehat{R}(g_k) - R(g_k)\big| \ge t\Big) \;\le\; 2\exp\big(-2m't^2/B^2\big),$$

so with $t = B\sqrt{\log(2K/\delta)/(2m')}$ and a union bound over $k \le K$, all K deviations are below t simultaneously with probability at least $1-\delta$. On that event, if k^* attains $\min_k R(g_k)$,

$$R(g_{\widehat{k}}) \;\le\; \widehat{R}(g_{\widehat{k}}) + t \;\le\; \widehat{R}(g_{k^*}) + t \;\le\; R(g_{k^*}) + 2t,$$

and $2t = B\sqrt{2\log(2K/\delta)/m'}$. □

S8.5 Proof of Proposition S1.5

Fix a coordinate j and abbreviate $\varphi = \varphi_j$, $Y(u) = G(u)_j$, $D = \sqrt{M+1}$. Each entry of $\varphi(u)$ is bounded by b in absolute value (the members by assumption, the intercept entry by construction), so $\|\varphi(u)\|_2 \le bD$, and for $\|w\|_2 \le W$ Cauchy–Schwarz gives $|w^\top\varphi(u)| \le WbD$ and $|w^\top\varphi(u) - Y(u)| \le b(1+WD)$. The losses $\ell_w(u) = (w^\top\varphi(u) - Y(u))^2$ therefore take values in $[0, L_\infty]$ with $L_\infty = b^2(1+WD)^2$.

Uniform deviation. Consider the one-sided suprema $Z^+ = \sup_{\|w\|_2 \le W}\big(\widehat{R}_j(w) - R_j(w)\big)$ and $Z^- = \sup_{\|w\|_2 \le W}\big(R_j(w) - \widehat{R}_j(w)\big)$ over the validation sample of size m. Changing one sample moves either supremum by at most L_∞/m, so McDiarmid's inequality gives $Z^\pm \le \mathbb{E}Z^\pm + L_\infty\sqrt{\log(4q/\delta)/(2m)}$, each with probability at least $1-\delta/(2q)$. By symmetrization,

$\mathbb{E}Z^{\pm} \le 2\,\mathfrak{R}_m(\mathcal{L})$, the Rademacher complexity of the loss class. The loss is the composition of $t \mapsto t^2$, restricted to $|t| \le b(1+WD)$ where it is $2b(1+WD)$-Lipschitz, with the class $\{u \mapsto w^\top \varphi(u) - Y(u)\}$; translation by the fixed function Y leaves the Rademacher average unchanged (the translated term does not involve w and has zero mean), and for the linear class over the ℓ^2 ball,

$$\mathfrak{R}_m \;\le\; \frac{W}{m}\,\mathbb{E}\Big\|\sum_{i=1}^m \sigma_i \varphi(\tilde u_i)\Big\|_2 \;\le\; \frac{W}{m}\Big(\sum_{i=1}^m \mathbb{E}\|\varphi(\tilde u_i)\|_2^2\Big)^{1/2} \;\le\; \frac{WbD}{\sqrt{m}},$$

by Jensen's inequality. Talagrand's contraction lemma then gives $\mathfrak{R}_m(\mathcal{L}) \le 2b(1+WD)\cdot WbD/\sqrt{m}$, hence

$$Z^{\pm} \;\le\; \frac{4Wb^2D(1+WD)}{\sqrt{m}} + L_\infty\sqrt{\frac{\log(4q/\delta)}{2m}} \qquad \text{jointly with probability} \ge 1-\delta/q.$$

Empirical risk minimization. On this event, for any minimizer w^* of R_j over the ball, $R_j(\widehat w_j) \le \widehat R_j(\widehat w_j) + Z^- \le \widehat R_j(w^*) + Z^- \le R_j(w^*) + Z^+ + Z^-$, and $Z^+ + Z^-$ is bounded by the displayed excess. A union bound over the two tails of each of the q coordinates makes all events hold simultaneously with probability at least $1-\delta$, and averaging over j preserves the (coordinate-uniform) bound. This proves (i).

For (ii), the ridge minimizer satisfies $\widehat R_j(\widehat w_j) + \lambda\|\widehat w_j\|_2^2 \le \widehat R_j(w^*) + \lambda\|w^*\|_2^2$, so $\widehat R_j(\widehat w_j) \le \widehat R_j(w^*) + \lambda\|w^*\|_2^2 \le \widehat R_j(w^*) + \lambda W^2$. Since by assumption $\|\widehat w_j\|_2 \le W$, the uniform event of part (i) applies to $\widehat w_j$ and w^* alike, and the preceding inequalities hold with the additional term λW^2. □

S8.6 Proof of Corollary 6.4

The lower bound is Corollary 6.3. For the upper bound, evaluate the quadratic form at the uniform weights $w_m = 1/M$:

$$\frac{1}{\bar e^{\,2}}\,w^\top S w = \frac{1}{\bar e^{\,2}}\Big(\frac{1}{M^2}\sum_m S_{mm} + \frac{1}{M^2}\sum_{m\neq k} S_{mk}\Big) \;\le\; \frac{1+\delta_e}{M} + \frac{M-1}{M}\,\big(\bar\varrho + \delta_\varrho\big),$$

using the entrywise upper bounds and the counts M and $M(M-1)$ of the two sums. The right side equals $\bar\varrho + \delta_\varrho + \big(1+\delta_e - \bar\varrho - \delta_\varrho\big)/M$, and the minimum over the simplex is at most the value at any feasible point. □

S8.7 Proof of Corollary S1.6

Fix a coordinate j and a grid point γ. Substituting the affine form of the correction,

$$h_{w,\gamma}(u)_j = w_j^\top \widetilde\varphi_{j,\gamma}(u) + o_{j,\gamma}(u), \qquad \widetilde\varphi_{j,\gamma}(u) = \varphi_j(u) - \Phi_j^\top c_\gamma(u), \quad o_{j,\gamma}(u) = c_\gamma(u)^\top Y_j,$$

and neither $\widetilde\varphi_{j,\gamma}$ nor $o_{j,\gamma}$ depends on w or on the validation sample: c_γ is built from the training design alone. Each component of $\Phi_j^\top c_\gamma(u)$ is a c_γ-weighted sum of member values bounded by b, so its absolute value is at most $b\,\|c_\gamma(u)\|_1 \le b\kappa$, whence $\|\widetilde\varphi_{j,\gamma}(u)\|_2 \le bD + bD\kappa = bD(1+\kappa)$ and $|o_{j,\gamma}(u)| \le b\kappa$. For $\|w\|_2 \le W$ the prediction deviates from the target by at most $WbD(1+\kappa)+b\kappa+b \le b(1+WD)(1+\kappa) =: \tilde{c}$, so the squared losses lie in $[0, \tilde{c}^2]$ and are $2\tilde{c}$-Lipschitz in the prediction.

By the argument of Proposition S1.5, for the class indexed by w in the W-ball and fixed (j,γ), the Rademacher complexity of the affine part is at most $W\,bD(1+\kappa)/\sqrt{m}$, contraction multiplies it by $2\tilde{c}$, symmetrization doubles it, and McDiarmid at level $\delta/(2q|\Gamma|)$ adds $\tilde{c}^2\sqrt{\log(4q|\Gamma|/\delta)/(2m)}$. A union bound over the two tails of the $q|\Gamma|$ pairs makes, with probability at least $1-\delta$, each one-sided deviation $Z^{\pm}_{j,\gamma}$, defined as in the proof of Proposition S1.5, at most

$$\frac{4Wb^2D(1+WD)(1+\kappa)^2}{\sqrt{m}} + b^2(1+WD)^2(1+\kappa)^2\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2m}}.$$

On that event, write $\bar{R}(\gamma) = \frac{1}{q}\sum_j R_j(h_{\widehat{w},\gamma})$ and $\widehat{\bar{R}}(\gamma)$ for its empirical version; averaging the $Z_{j,\gamma}$ over j bounds $|\widehat{\bar{R}}(\gamma) - \bar{R}(\gamma)|$ by the same quantity for every γ simultaneously, including at the data-dependent $\widehat{w}$, because the deviations are uniform over the ball. The selection chain $\bar{R}(\widehat{\gamma}) \le \widehat{\bar{R}}(\widehat{\gamma}) + Z \le \widehat{\bar{R}}(\gamma^*) + Z \le \bar{R}(\gamma^*) + 2Z$ with γ^* the aggregate oracle then gives the display, whose constant is $2Z$ in the factored form stated. When the c_γ are kernel ridge weights, Lemma S1.7 supplies the hypothesis with $\kappa = \kappa_{\mathrm{an}}$. $\square$

S8.8 Proof of Lemma S1.7

Fix u and write $k_u = k(X,u)$, $t = n\lambda$, $A = K + tI$ and $c = A^{-1}k_u$. Because k is positive definite, the Gram matrix of the $n+1$ points $u_1, \dots, u_n, u$ is positive semidefinite, and by the generalized Schur complement this places k_u in the range of K with $k_u^\top K^+ k_u \le k(u,u)$, K^+ the pseudo-inverse. Diagonalize $K = \sum_i \mu_i v_i v_i^\top$ and expand $k_u = \sum_i \beta_i v_i$ over the eigenvectors with $\mu_i > 0$; the Schur condition reads $\sum_i \beta_i^2/\mu_i \le k(u,u)$. Then

$$\|c\|_2^2 = \sum_i \frac{\beta_i^2}{(\mu_i+t)^2} = \sum_i \frac{\beta_i^2}{\mu_i}\cdot\frac{\mu_i}{(\mu_i+t)^2} \;\le\; \frac{1}{4t}\sum_i \frac{\beta_i^2}{\mu_i} \;\le\; \frac{k(u,u)}{4t},$$

because $(\mu+t)^2 \ge 4\mu t$ for all $\mu, t \ge 0$. Cauchy–Schwarz gives $\|c\|_1 \le \sqrt{n}\,\|c\|_2$, and combining, $\|c\|_1^2 \le n\|c\|_2^2 \le k(u,u)/(4\lambda)$.

For the sharpness take $k(x,y) = \cos(x-y)$, a positive definite kernel on the line (its native space is spanned by cos and sin), the two points $x_1 = 0$ and $x_2 = \theta$ with $\theta \in (0,\pi)$ chosen so that $1-\cos\theta = 2\lambda$, and $u = (\theta+\pi)/2$. Then $K = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1\end{smallmatrix}\right)$ has the eigenvector $v = (1,-1)/\sqrt{2}$ with eigenvalue $\mu = 1-\cos\theta = t$, and $k_u = (\cos u, \cos(u-\theta)) = -\sin(\theta/2)\,(1,-1)$ is proportional to v, with $\beta^2/\mu = 2\sin^2(\theta/2)/(1-\cos\theta) = 1 = k(u,u)$. Hence $c = A^{-1}k_u = k_u/(2t)$ has entries of equal magnitude, $\|c\|_2^2 = \beta^2/(2t)^2 = 1/(4t)$, and $\|c\|_1 = \sqrt{2}\,\|c\|_2 = \sqrt{n}\,\|c\|_2 = \frac{1}{2}\lambda^{-1/2}$, so every inequality in the display is an equality.

For the Matérn kernel of Section 3.3, $k(u,u) = 1$ and the smallest nugget on the grid is 10^{-6}, so $\sup_{u,\gamma}\|c_\gamma(u)\|_1 \le \frac{1}{2}\max_\gamma \lambda_\gamma^{-1/2} = 500$ irrespective of the length scale and of the design. $\square$

S8.9 Proof of Corollary 6.5

The objective is strictly convex on the hyperplane $\{\mathbf{1}^\top w = 1\}$, so its minimizer there is the unique stationary point of the Lagrangian $w^\top S w - \nu(\mathbf{1}^\top w - 1)$, namely $w = \frac{\nu}{2}S^{-1}\mathbf{1}$ with

ν fixed by the constraint: $w^\star = S^{-1}\mathbf{1}/(\mathbf{1}^\top S^{-1}\mathbf{1})$, at which $w^{\star\top} S w^\star = 1/(\mathbf{1}^\top S^{-1}\mathbf{1})$. The simplex is a subset of the hyperplane, so the convex minimum is at least this value, and by uniqueness of the minimizer the two agree exactly when $w^\star$ lies in the simplex, that is, has no negative coordinate. $\square$

S8.10 Proof of Proposition 6.1

Since $\sum_m w_m = 1$, the stack's normalized residual is $\rho_{f_w}(u) = \sum_m w_m \rho_{f_m}(u)$, hence

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 = \mathbb{E}\Big\| \sum_m w_m \rho_{f_m}(u)\Big\|_2^2 = \sum_{m,k} w_m w_k\, \mathbb{E}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle = w^\top S w.$$

For (i), parameterize $w = (1-t, t)$; $q(t) = w^\top S w$ is a convex quadratic in t with $q'(0) = 2(\varrho e_1 e_2 - e_1^2)$, negative precisely when $\varrho < e_1/e_2$. In that case the unconstrained minimizer lies in $(0,1)$ and gives the stated interior value; otherwise $q'(0) \ge 0$, the minimum over $[0,1]$ is at $t = 0$ with value e_1^2, and the second member is dropped. For (ii), $S = e^2\big((1-\varrho)I + \varrho\,\mathbf{1}\mathbf{1}^\top\big)$, so $w^\top S w = e^2\big((1-\varrho)\|w\|_2^2 + \varrho\big)$ on the simplex, minimized by the uniform weights, where $\|w\|_2^2 = 1/M$. $\square$

S8.11 Proof of Corollary 6.3

Convex weights are nonnegative, so each term of $w^\top S w = \sum_m w_m^2 S_{mm} + \sum_{m\neq k} w_m w_k S_{mk}$ can be bounded below entrywise:

$$w^\top S w \;\ge\; \bar{e}^{\,2} \sum_m w_m^2 + \bar{\varrho}\,\bar{e}^{\,2} \sum_{m\neq k} w_m w_k = \bar{e}^{\,2}\Big(\|w\|_2^2 + \bar{\varrho}\,(1 - \|w\|_2^2)\Big),$$

using $\sum_{m\neq k} w_m w_k = (\sum_m w_m)^2 - \|w\|_2^2 = 1 - \|w\|_2^2$. The bracket is a convex combination of 1 and $\bar{\varrho}$, hence at least $\bar{\varrho}$. $\square$

S8.12 Proof of Theorem 6.6

Write $W_a = \sum_k w_{a,k}$ for the total weight of architecture a, so that $\sum_a W_a = 1$. Split the quadratic form into its diagonal terms, the terms between two seeds of one architecture and the terms between architectures, and bound each entry below by its hypothesis; the weights are nonnegative, so multiplying by the weights preserves these inequalities:

$$w^\top S w \;\ge\; \bar{e}^{\,2}\Big(\|w\|_2^2 + \varrho_w\big(\textstyle\sum_a W_a^2 - \|w\|_2^2\big) + \varrho_b\big(1 - \sum_a W_a^2\big)\Big),$$

using $\sum_{k\neq l} w_{a,k} w_{a,l} = W_a^2 - \sum_k w_{a,k}^2$ within an architecture and $\sum_{a\neq b} W_a W_b = 1 - \sum_a W_a^2$ across them. Rearranged, the right side is $\bar{e}^{\,2}\big(\varrho_b + (\varrho_w - \varrho_b)\sum_a W_a^2 + (1-\varrho_w)\|w\|_2^2\big)$. Both coefficients are nonnegative by hypothesis; $\sum_a W_a^2 \ge 1/M$ by Cauchy–Schwarz, with equality at $W_a = 1/M$; and $\|w\|_2^2 \ge 1/(MK)$ with equality at uniform weights, which also give $W_a = 1/M$. So the minimum of the right side over the simplex is attained at uniform weights and equals the display. When the three hypotheses hold with equality, $S = \bar{e}^{\,2}\big((1-\varrho_w)I + (\varrho_w - \varrho_b)B + \varrho_b\mathbf{1}\mathbf{1}^\top\big)$ with B the indicator of a shared architecture, the first inequality is an identity for every w, and uniform weights attain the bound. The three limits follow by letting K, then M, grow. $\square$

S8.13 Proof of Proposition 6.7

Proposition 6.1(i) gives $V = e_1^2 e_2^2(1-\varrho^2)/(e_1^2+e_2^2-2\varrho e_1 e_2) = e_1^2\,t^2(1-\varrho^2)/D$ with $D = 1+t^2-2\varrho t$, valid whenever the unconstrained minimizer lies in the open simplex, which is the condition $\varrho < 1/t$ for $t = e_2/e_1 \geq 1$. The strict derivative claims additionally assume $-1 < \varrho$, as in the proposition. Differentiating,

$$\begin{aligned}\frac{\partial V}{\partial t} &= \frac{e_1^2\big[2t(1-\varrho^2)D - t^2(1-\varrho^2)(2t-2\varrho)\big]}{D^2}\\ &= \frac{2e_1^2\,t(1-\varrho^2)\big(D - t^2 + \varrho t\big)}{D^2}\\ &= \frac{2e_1^2\,t(1-\varrho^2)(1-\varrho t)}{D^2},\end{aligned}$$

and

$$\begin{aligned}\frac{\partial V}{\partial \varrho} &= \frac{e_1^2\big[-2\varrho t^2 D + 2t^3(1-\varrho^2)\big]}{D^2}\\ &= \frac{2e_1^2\,t^2\big(t - \varrho - \varrho t^2 + \varrho^2 t\big)}{D^2}\\ &= \frac{2e_1^2\,t^2(t-\varrho)(1-\varrho t)}{D^2}.\end{aligned}$$

Both are positive for $-1 < \varrho < 1/t$, since $1-\varrho^2 > 0$, $t-\varrho > 0$ and $1-\varrho t > 0$ there. Setting $\mathrm{d}V = (\partial V/\partial t)\,\mathrm{d}t + (\partial V/\partial \varrho)\,\mathrm{d}\varrho = 0$ and cancelling the common factor $2e_1^2 t(1-\varrho t)/D^2$ gives $(1-\varrho^2)\,\mathrm{d}t + t(t-\varrho)\,\mathrm{d}\varrho = 0$, which is the stated rate; substituting $\mathrm{d}t = -\epsilon t$ gives the fractional form. □

S8.14 Proof of Proposition S2.2

The metric is diagonal, so the squared norm splits over coordinates and the j-th summand $w_j^2\,\mathbb{E}\big(\sum_m v_{mj} f_m(u)_j - G(u)_j\big)^2$ depends on the weights only through the column $v_{\cdot j}$. Minimizing the sum is therefore q independent minimizations, and the positive factor w_j^2 leaves every coordinate minimizer unchanged, so the optimizer is the same for every positive w. A combination with shared weights is the special case $v_{mj} = v_m$ for all j, a subset of the feasible set of each coordinate problem, which gives the domination. □

S8.15 Proof of Corollary S2.3

The integrand is nonnegative, so Tonelli's theorem permits exchanging the expectation with the coordinate sum:

$$\mathbb{E}\Big[a(u)\sum_j w_j^2\big(f_v(u)_j - G(u)_j\big)^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\big(f_v(u)_j - G(u)_j\big)^2\Big].$$

The j-th summand depends on v only through its j-th column $v_{\cdot j}$, exactly as in the proof of Proposition S2.2, so minimizing over v splits into q independent problems; the factors w_j^2 rescale the summands without moving any per-coordinate minimizer, which gives the simultaneous optimality over all diagonal w. The choice $a(u) = \|G(u)\|_2^{-2}$, $w = \mathbf{1}$ identifies the objective with $\mathbb{E}\,\|\rho_{f_v}\|_2^2$, the mean squared relative error. Finally $\mathbb{E}\|\rho\|_2 \leq (\mathbb{E}\|\rho\|_2^2)^{1/2}$ is Jensen's inequality. □

S8.16 Proof of Proposition S2.4

Fix a coordinate j and a member m. The validation scores $a(\tilde{u}_i)(f_m(\tilde{u}_i)_j - G(\tilde{u}_i)_j)^2$, $i \le m_v$, are i.i.d., take values in $[0, L]$ by assumption, and have mean $R_j(m)$; the members and the weight are fixed relative to the validation sample. Hoeffding's inequality gives

$$\Pr\Big(\big|\widehat{R}_j(m) - R_j(m)\big| \ge t\Big) \;\le\; 2\exp\big(-2m_v t^2/L^2\big)$$

for each of the Mq pairs (m, j). With $t = L\sqrt{\log(2Mq/\delta)/(2m_v)}$ and a union bound, all Mq deviations are below t simultaneously with probability at least $1-\delta$. On that event, for every j, writing $m^*(j)$ for a population minimizer,

$$R_j\big(\widehat{m}(j)\big) \le \widehat{R}_j\big(\widehat{m}(j)\big) + t \le \widehat{R}_j\big(m^*(j)\big) + t \le R_j\big(m^*(j)\big) + 2t.$$

The metric statement follows by multiplying the coordinate-wise inequality by $w_j^2 \ge 0$ and summing; the selection $\widehat{m}(\cdot)$ does not depend on w, so one event serves every diagonal metric. $\square$

S8.17 Proof of Proposition 6.8

Let $T : \mathcal{H}_k \to \mathbb{R}^{\mathcal{X}}$ be the composition map $Tf = f \circ \varphi$. For $x \in \mathcal{X}$ the reproducing property gives $(Tf)(x) = f(\varphi(x)) = \langle f, k(\cdot, \varphi(x))\rangle_{\mathcal{H}_k}$, so evaluation of Tf at x is a bounded functional. The kernel of T is the intersection of the kernels of these bounded evaluations and is closed. Its orthogonal complement therefore contains a unique minimum-norm preimage of every element in the image. Thus the image $\mathcal{H}_{k_\varphi} := T(\mathcal{H}_k)$ carries the quotient norm $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : Tf = g\}$, the minimum over the affine subspace $T^{-1}(g)$ (nonempty exactly when g is a pullback). The quotient by this closed subspace is complete. It is an RKHS, with reproducing kernel $k_\varphi(x, x') = \langle k(\cdot, \varphi(x)), k(\cdot, \varphi(x'))\rangle_{\mathcal{H}_k} = k(\varphi(x), \varphi(x'))$, since the minimum-norm representer of evaluation at x is $k(\cdot, \varphi(x))$. Taking $f = h$ in the minimum gives $\|h \circ \varphi\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and substituting this bound into Theorem 6.9 applied with the kernel k_φ replaces $\|G\|$ by $\|h\|_{\mathcal{H}_k}$. $\square$

S8.18 Proof of Proposition S2.5

The truncated singular-value decomposition satisfies $\|A - A_r\|_{\mathrm{op}} = \sigma_{r+1}$. Consequently, for every $u \in \mathcal{U}$,

$$\|G(u) - G_r(u)\|_2 = \|g(Au) - g(A_r u)\|_2 \le L\|(A - A_r)u\|_2 \le LR\sigma_{r+1} = \varepsilon.$$

Write $\delta(u) = G(u) - G_r(u)$, and form the kernel predictor of G_r using the same feature-space weights $c = c_\lambda(u)$. Theorem 6.9 applied in feature space with zero mean gives

$$\|G_r(u) - c^\top G_r(X)\|_2 = \|h(\varphi_r(u)) - c^\top h(Z)\|_2 \le \|h\|_K\, \widetilde{P}_\lambda(u).$$

The estimator in the proposition is trained on $G(X)$, so

$$G(u) - c^\top G(X) = G_r(u) - c^\top G_r(X) + \delta(u) - \sum_i c_i \delta(u_i).$$

Taking Euclidean norms and using $\|\delta(u_i)\|_2 \le \varepsilon$ yields the stated bound. The argument is deterministic and applies to any design and any stated nugget, including the pseudoinverse convention at zero. $\square$

S8.19 Proof of Theorem 6.9

Fix u and write $k_u = k(X,u) \in \mathbb{R}^n$. At $\lambda = 0$, interpret the inverse as K^+ if K is singular. For $z \in \ker K$, the RKHS function $h_z = \sum_i z_i k(u_i, \cdot)$ has squared norm $z^\top K z = 0$. Hence $z^\top k_u = \langle h_z, k(u,\cdot)\rangle = 0$, which places k_u in range K. In particular, $KK^+ k_u = k_u$, so the algebra below also applies under the pseudoinverse convention.

Put $A = K + n\lambda I$ and $w = A^{-1}k_u$. For each component j, membership $r_j \in \mathcal{H}_k$ and the reproducing property give

$$r_j(u) - \widehat{r}_{\lambda,j}(u) = \Big\langle r_j,\ k(u,\cdot) - \sum_{i=1}^{n} w_i\, k(u_i,\cdot) \Big\rangle_{\mathcal{H}_k},$$

because $\widehat{r}_{\lambda,j}(u) = k_u^\top A^{-1} r_j(X) = \sum_i w_i\, r_j(u_i)$ and $r_j(u_i) = \langle r_j, k(u_i,\cdot)\rangle$. Cauchy–Schwarz yields

$$\big|r_j(u) - \widehat{r}_{\lambda,j}(u)\big| \ \le\ \|r_j\|_{\mathcal{H}_k} \Big\| k(u,\cdot) - \sum_i w_i k(u_i,\cdot)\Big\|_{\mathcal{H}_k},$$

and the second factor is independent of j; call it $\rho(u)$. Expanding,

$$\rho(u)^2 = k(u,u) - 2\,w^\top k_u + w^\top K w = k(u,u) - k_u^\top A^{-1}\big(2A - K\big)A^{-1}k_u.$$

Since $2A - K = A + n\lambda I$,

$$\rho(u)^2 = k(u,u) - k_u^\top A^{-1}k_u - n\lambda\,\|A^{-1}k_u\|_2^2 = \widetilde{P}_\lambda(u)^2 \ \le\ P_\lambda(u)^2.$$

Summing the squared componentwise bounds,

$$\big\|r(u) - \widehat{r}_\lambda(u)\big\|_2^2 = \sum_j \big(r_j(u) - \widehat{r}_{\lambda,j}(u)\big)^2 \le \rho(u)^2 \sum_j \|r_j\|_{\mathcal{H}_k}^2 = \widetilde{P}_\lambda(u)^2\, \|r\|_K^2.$$

For the sharpness, write $g_u = k(u,\cdot) - \sum_i w_i k(u_i,\cdot)$, so that $\|g_u\|_{\mathcal{H}_k} = \rho(u) = \widetilde{P}_\lambda(u)$. If $\widetilde{P}_\lambda(u) = 0$ both sides of the bound vanish for every r. Otherwise take $r_1 = (\|G - m\|_K/\rho(u))\, g_u$ and $r_j = 0$ for $j \ge 2$: this residual has norm $\|G - m\|_K$, and since the estimator is built from its own training values $r_1(X) = (\|G-m\|_K/\rho(u))g_u(X)$, the first display gives $r_1(u) - \widehat{r}_{\lambda,1}(u) = \langle r_1, g_u\rangle_{\mathcal{H}_k} = \|G-m\|_K\,\rho(u)$, equality in Cauchy–Schwarz. $\square$

The argument applies for every $\lambda \ge 0$. The inequality between $\widetilde{P}_\lambda$ and P_λ compares the two factors at the same ridge value; Theorem 6.10 describes how that value affects the worst-case error.

S8.20 Proof of Theorem 6.10

Write $k_u = k(X,u)$ and let $g_u = k(u,\cdot) - \sum_i (K^{-1}k_u)_i k(u_i,\cdot)$ be the function of the previous proof at $\lambda = 0$. It vanishes on the design, since $g_u(u_j) = k(u,u_j) - k_u^\top K^{-1}Ke_j = 0$, and $g_u(u) = \|g_u\|_{\mathcal{H}_k}^2 = k(u,u) - k_u^\top K^{-1}k_u = P_0(u)^2$. If $P_0(u) = 0$ the lower bound is trivial. Otherwise set $r^\pm = \pm(\rho/P_0(u))\, g_u\, e_1$, two residual operators of norm ρ whose training values are both zero, so that any Ψ returns the same vector $\psi = \Psi(0)$ for both; their values at u are $\pm\rho P_0(u)e_1$, hence $\max_\pm \|r^\pm(u) - \psi\|_2 \ge \max_\pm |\pm\rho P_0(u) - \psi_1| \ge \rho P_0(u)$. This is the lower bound. Theorem 6.9 with $\lambda = 0$ bounds the interpolant's error by $\|r\|_K P_0(u) \le \rho P_0(u)$ on the whole ball, so the interpolant attains it, and for $\lambda > 0$ the same theorem together with its sharpness case gives the ridge correction's worst case as exactly $\rho\,\widetilde{P}_\lambda(u)$.

For the identity, put $t = n\lambda$ and $A = K + tI$, which commutes with K. From the proof of Theorem 6.9, $\widetilde{P}_\lambda(u)^2 = k(u,u) - k_u^\top A^{-1}(2A-K)A^{-1}k_u$, while $P_0(u)^2 = k(u,u) - k_u^\top K^{-1}k_u$, so

$$\begin{aligned}\widetilde{P}_\lambda(u)^2 - P_0(u)^2 &= k_u^\top\big(K^{-1} - 2A^{-1} + A^{-1}KA^{-1}\big)k_u \\ &= k_u^\top A^{-2}K^{-1}\big(A^2 - 2AK + K^2\big)k_u \\ &= k_u^\top A^{-2}K^{-1}(A-K)^2 k_u,\end{aligned}$$

and $A - K = tI$. The right-hand side is nonnegative because $A^{-2}K^{-1}$ is positive definite, which gives $P_0 \le \widetilde{P}_\lambda$; the upper inequality $\widetilde{P}_\lambda \le P_\lambda$ was shown above. Expanding k_u in the eigenbasis of K as in the proof of Lemma S1.7, the difference equals $\sum_i(\beta_i^2/\mu_i)\,(t/(\mu_i+t))^2$, each term of which is bounded by β_i^2/μ_i and tends to zero with t, so the difference vanishes as $\lambda \to 0$ by dominated convergence. □

S8.21 Proof of Proposition S3.1

Condition on the fixed fitted predictor, scale function and design. The calibration scores $s_1, \dots, s_m$ and the new score $s^\star$ are exchangeable, and coverage is the event $s^\star \le Q$. If $k = m+1$ then $Q = +\infty$ and coverage is one. Suppose therefore that $k \le m$.

Assign independent continuous random tie breakers to the $m+1$ scores and let J be the rank of the new score in this augmented ordering. By exchangeability, J is uniform on $\{1, \dots, m+1\}$. The event $J \le k$ implies $s^\star \le s_{(k)}$, including in the presence of ties, so

$$\mathbb{P}(s^\star \le Q) \ge \mathbb{P}(J \le k) = \frac{k}{m+1} \ge 1 - \alpha.$$

When the scores are almost surely distinct the two events coincide; their probability equals $k/(m+1) \le 1-\alpha+1/(m+1)$. The same upper bound also includes the already handled case $k = m+1$. Averaging over any independent fitted objects preserves these statements. □

S8.22 Proof of Corollary S5.2

The nonzero eigenvalues of $K/n = XX^\top/n$ coincide with those of the sample covariance matrix $S = X^\top X/n \in \mathbb{R}^{p\times p}$, and zero eigenvalues contribute nothing to $d_{\mathrm{eff}}(\lambda) = \sum_i \lambda_i(K)/(\lambda_i(K)+n\lambda)$, so

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} = \frac{p}{n}\cdot\frac{1}{p}\sum_{j=1}^{p}\frac{\lambda_j(S)}{\lambda_j(S)+\lambda} = \frac{p}{n}\Big(1 - \lambda\,\widehat{m}_S(-\lambda)\Big),$$

by the computation of Lemma S5.1 applied to S, with $\widehat{m}_S$ the Stieltjes transform of the empirical spectral distribution of S. By the Marčenko–Pastur theorem (Marčenko and Pastur, 1967), this distribution converges weakly, almost surely, to the law with ratio γ and scale σ^2, and since $x \mapsto (x+\lambda)^{-1}$ is bounded and continuous on $[0,\infty)$, $\widehat{m}_S(-\lambda) \to m(-\lambda)$. It remains to evaluate $m(-\lambda)$. The Stieltjes transform of the Marčenko–Pastur law satisfies the self-consistent equation

$$m(z) = \frac{1}{\sigma^2\big(1-\gamma-\gamma z\,m(z)\big) - z},$$

i.e. $\gamma\sigma^2 z\, m^2 + \big(z - \sigma^2(1-\gamma)\big)m + 1 = 0$. At $z = -\lambda$ the discriminant is $\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda > 0$, and of the two real roots

$$m(-\lambda) = \frac{\sigma^2(1-\gamma) + \lambda \pm \sqrt{\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda}}{-2\gamma\sigma^2\lambda}$$

only the one with the minus sign in the numerator is positive, as $m(-\lambda) = \int (x+\lambda)^{-1} dF(x) > 0$ requires; rearranging gives the stated expression. $\square$

S8.23 Proof of Lemma S5.1

Diagonalize K with eigenvalues λ_i. Then

$$\begin{aligned}
\operatorname{tr}\big(K(K+n\lambda I)^{-1}\big) &= \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_i + n\lambda} \\
&= \sum_{i=1}^{n} \Big(1 - \frac{n\lambda}{\lambda_i + n\lambda}\Big) \\
&= n - \lambda \sum_{i=1}^{n} \frac{1}{\lambda_i/n + \lambda} \\
&= n\big(1 - \lambda\, \widehat{m}(-\lambda)\big),
\end{aligned}$$

using $\widehat{m}(-\lambda) = \frac{1}{n}\sum_i (\lambda_i/n + \lambda)^{-1}$. The limit statement follows from the weak convergence of the empirical spectral distribution together with the boundedness and continuity of $x \mapsto (x+\lambda)^{-1}$ on $[0,\infty)$ for fixed $\lambda > 0$. $\square$

S9 Implementation details

The structural-mechanics experiments used CPU-only cloud instances, with single-precision network training and double-precision kernel solves and error computations. The second experiment of Section 4.1 ran on one Intel Xeon E5-2698 v4 host (20 physical cores, 40 hardware threads). The distributed `StructuralMechanics` dataset contains 40000 samples. Its inputs have an exact broadcast structure and are represented in $\mathbb{R}^{41}$. Samples 1–20000 form the training block and samples 20001–40000 form the test block. In the high-data protocol, a fixed permutation assigns 1000 training-block samples to validation and 19000 to fitting. The low-data protocol uses the first 1250 training-block samples, with the final 250 reserved for validation. Model selection uses the validation split.

FNO. The network has width 64, 14 modes per dimension, and 4 spectral layers with pointwise linear skips and GELU activations. The input lift takes 3 channels (the broadcast load and two coordinates), and the output projection has dimensions $64 \to 128 \to 1$, giving 6.45M parameters in total. Training uses AdamW with learning rate 1.5×10^{-3}, weight decay 10^{-6}, a cosine schedule ending at 10^{-6}, and batch size 128. The first experiment scheduled 200 epochs and retained the best validation checkpoint reached before its time limit; the second completed a 100-epoch schedule.

Transformer. This architecture is excluded from the reported ensemble. Its encoder uses 41 tokens, each comprising the load value and 10 sine/cosine pairs of the coordinate, with 5 pre-norm blocks, dimension 192, 4 attention heads, and MLP ratio 4. The decoder forms 1681 grid queries from Fourier features of both coordinates and uses two cross-attention blocks followed by a pointwise $192 \to 192 \to 1$ head. The network has 3.16M parameters.

UNet. The network has three convolutional scales ($41 \to 21 \to 11$ by average pooling with ceiling mode) and a 6×6 bottleneck. Each block uses two 3×3 convolutions with GroupNorm(8) and SiLU, with widths 48/96/192/384. Bilinear upsampling, skip concatenation, and a 1×1 output head complete the network, which has 4.38M parameters and the same input channels as the FNO. Training uses AdamW with learning rate 1.5×10^{-3}, weight decay 10^{-5}, a cosine schedule ending at 10^{-6}, and batch size 256. The second experiment uses 100 epochs.

MSE variant. The residual MLP uses mean squared error on standardized targets in place of the metric loss, with a 120-epoch schedule and otherwise the same training settings. It has the lowest error among the plain MLPs and introduces a second training loss into the ensemble.

MLP. The input layer has dimensions $41 \to 1024$, followed by three residual SiLU blocks of width 1024 and an output layer $1024 \to 1681$, for 4.91M parameters. Training uses AdamW with learning rate 10^{-3}, weight decay 10^{-5}, a cosine schedule, 400 epochs, and batch size 256. A wider variant has width 1536, five residual blocks and 14.5M parameters, with the same training protocol. The low-data pipeline uses the width-1024 MLP for 400 epochs with batch size 64 on the fixed 1000 fitting rows.

Refiner. The load is concatenated with its kernel stress prediction and passed through an input layer of dimensions $(41 + 1681) \to 1024$ and the same residual trunk as the MLP. The resulting network has 6.64M parameters and uses the MLP training settings for 100 epochs. Its conditioning channel consists of four-fold out-of-fold kernel predictions during training and full-data kernel predictions at evaluation. Reflection augmentation flips the load and permutes both the kernel channel and the target by the row-reversal index. Evaluation averages $F(u, h(u))$ with $TF(Su, Th(u))$; it does not recompute the kernel prediction at Su. The full-map guarantee and the additional condition on the kernel channel are distinguished in Supplement S1.1.

Common training details. Except for the MSE variant, the loss is the batch mean of $\|\hat{v} - v\|_2/\|v\|_2$ in original units. Targets are standardized by the per-pixel training mean and global standard deviation, and predictions are denormalized when evaluating the loss. Neural members use reflection augmentation with probability 1/2 and reflection averaging at evaluation, with the supplied-field convention for the refiner described above. The kernel-only member uses neither operation. Validation error is evaluated every 10 epochs and the checkpoint with the lowest error is retained. Section 4.6 compares residual alignment within and between configurations in the sixty-predictor pool.

Test-block use. The structural-mechanics test block is the benchmark's fixed public split. It is used by the first structural-mechanics experiment, the six-member experiment, the ablations, the hard-case study, the learning curves, the sixty-predictor analysis, the conformal calibration (which draws its 1000 calibration cases from it) and the additional controls. Predictor fitting and model-level selection use the fitting and validation splits. The test-block analyses additionally fit the hindsight ensemble optimum and choose calibration quantiles on a test subset. The same 2000 public OCO-2 test states are reused across the ten-seed experiment, the readout control and the ensemble comparison.

Training labels and target centering. The pooled-centering runs use `krr_oof.py` with `--target-centering pooled`; the fold-local variant uses `--target-centering fold-local`. Only the kernel channel uses foldwise fitted kernel weights. Its predictions use a pooled target mean over all 19000 fitting rows, so held-out fold targets enter that centering statistic. Neural training predictions are generated by the networks fitted on those rows, and the refiner uses its training kernel channel. The correction is therefore fitted to $Y - M_{\mathrm{tr}}$. At query time the mean uses full-data kernel predictions. Section 6.4 gives the explicit $m_{\mathrm{full}}(X) - M_{\mathrm{tr}}$ mismatch term. Supplement S11 measures its propagation with the fitted mean and kernel fixed, and compares target-centering choices after repeating refiner training and downstream fitting.

Seed protocol. In the high-data experiment, one seed value enters the stochastic components: the permutation that draws the validation split from the training pool, each member's initialization and batch order, the half-split of the stacking comparison, the subsamples used to tune the kernel stages, and the conformal calibration draw; the pool/test boundary remains fixed. In the low-data protocol the first 1000 fitting and next 250 validation rows remain fixed across seeds. ClimSim provides no fixed test split, so its seed also draws the 20000-row test block out of the full set (`campaign/climsim_seeded.py`). ClimSim seed variation therefore includes test sampling, whereas the structural-mechanics and OCO-2 test blocks are fixed. Tables report the mean and standard deviation across seeds.

Calibration and readout controls. The conformal comparison uses the same calibration split for the constant-radius, disagreement-scaled and P_λ-scaled bands. Each seed's P_λ is computed from its selected correction kernel (Section 4.7). The OCO-2 readout control repeats network training at ten seeds per band and fits ridge and kernel readouts to the same frozen features within each repetition (Table 7). The ClimSim kernel-budget comparison varies the number of fitting rows under the protocol of Supplement S7.

Cost of individual kernel stages. The parameter counts of Table 4 are the networks' weights alone. A deployed structural-mechanics pipeline also stores the coefficient matrix of the kernel member and that of the residual correction, each 19000×1681, about 31.9 million numbers apiece (255 MB each in double precision), and evaluates a kernel row against the 19000 training loads for every query. At these dimensions, an eight-thread calculation on the same host uses synthetic values to time the arithmetic in each operation. Block-filled Gram construction takes 27 s, Cholesky factorization takes 22 s, and triangular solves with 1681 right-hand sides take 11 s. The peak resident memory is 6.0 GiB against 19.2 GiB after full-matrix construction. The lower peak uses block-filled construction; the experimental correction implementation, `correct_stack.py`, uses full-size temporaries. At inference one kernel stage costs 14 ms for a single query and 1.7 ms per query in batches of a thousand. These timings measure one kernel stage. The full six-member pipeline evaluates two such stages plus five neural members with reflection averaging; its total inference latency was not measured by this diagnostic.

Experimental schedules. The first structural-mechanics experiment used ten seeds on four CPU instances. Kernel ridge, the metric-loss MLP, the normalized-MSE MLP and the refiner completed their schedules at every seed. A 36-hour per-task limit interrupted all ten FNO schedules. The reported FNO predictions use the best observed validation checkpoint at each seed, selected between epochs 10 and 70 of the scheduled 200. Validation error increased after the selected checkpoint in every observed training curve; the incomplete schedules do not establish performance after full training. No UNet was trained in this experiment. Four- and five-member stacks, residual corrections, second-moment measurements and conformal calibrations were completed at every seed. Kernel-stage timings distinguish the standard and larger host classes (Table S9.1); Section S6.4 compares the memory requirements of full-matrix and block-filled Gram construction.

The second experiment used the same ten seeds on one host with 20 physical cores and 40 hardware threads. Repeated fits of kernel ridge and the three MLP-family members agree with the first experiment's test errors to within 6×10^{-8} in the relative-error unit for the two MLPs and the kernel ridge and 6×10^{-6} for the refiner. The FNO and UNet each use a completed 100-epoch cosine schedule. Validation selects the final checkpoint for nine FNO runs and all ten UNet runs, and the epoch-79 checkpoint for the remaining FNO run. Each cosine schedule is normalized to 100 epochs and ends at 10^{-6}. The six-member analysis includes per-pixel and global convex stacking, residual correction, second moments and conformal calibration at every seed. The paired comparisons of Section 4.3 repeat these analyses on the five members excluding the UNet; the half-split comparison selects global convex weights at one seed of the five-member experiment, giving a corrected global convex stack. Table S9.2 gives the training costs of each member.

The OCO-2 raw-input kernel size sweep uses ten seeds per band. The separate O2 learning curve in Section S7.2 uses one seed; the ClimSim curves use five, three and three seeds at their respective training sizes. The low-data pipeline uses ten seeds. The OCO-2 head pipeline uses ten seeds on each of the three bands, giving thirty band-seeds at a matched 250-epoch budget, with one additional run at 750 epochs to examine sensitivity to training duration. The combination rules, empirical floors and ensembles in Section 5 use member predictions and per-sample errors from each band-seed. The readout control uses the same seeds and data-split protocol as Table 6; small differences in weighted-network errors reflect repeated training. Seven of the thirty band-seeds were also repeated on different hardware: six of the ten rows reproduce to the printed precision across the two runs, including the kernel rows, the flat-metric network, its feature head and the reported combination, while the weighted-loss networks and their kernel heads differ by up to 0.39 percentage points in reduced error. Numerical reduction order can vary across hardware despite fixed initialization and data order. These differences are smaller than the corresponding between-seed standard deviations.

Prediction precision. Errors recomputed from the stored member predictions use the same metric and seed-specific validation partitions as the original evaluations. On the fixed test block, all forty comparisons with the original reflection-averaged errors agree to between 10^{-14} and 6×10^{-8}, consistent with the storage precision.

Residual data and rounding. The residual and selection analyses of Sections 4.5 and 4.3 use $r_m = f_m - y$. For the first experiment, each of the ten seeds has one compressed NumPy archive, `sm_residuals_s`k`.npz`, containing the test-block residual of each of the five members and of the reported pipeline (per-pixel stack plus kernel correction) as a 20000×1681 array, the validation-split residual of each member as a 1000×1681 array, the target norms $\|y\|$ of every sample on both splits, and the index arrays of both splits; the residuals are stored in IEEE half precision (`float16`), the norms and indices at full width. Across all 110 arrays, the mean validation and test errors computed from the half-precision residuals differ from the original values by at most 5.4×10^{-8} in relative-error units. The largest stored residual is 192, compared with target norms of 3.7×10^3 to 1.7×10^4. The measured rounding effect on these mean errors is far below their seed spread. The archives occupy 384 MB per seed, 3.84 GB in all. The archive manifest specifies the member, seed, split, dimensions, precision, checksums and error summaries. The member errors in Table 4 are sample means of $\|r_m\|/\|y\|$. The normalized validation residuals $r_m/\|y\|$ determine the second-moment matrix S and simplex weights; the test risks are computed from $\|\sum_m w_m r_m\|/\|y\|$ using the sum-to-one identity. The same arrays supply the dispersion factor in Proposition 6.1, pairwise correlations, the equicorrelation summary, equal-weight and fitted ensemble errors, spatial error maps, and the hard-sample comparison in Section 4.5. Conformal scores use $\|r\|$ and the same seed-specific calibration subset of the test block. All these quantities inherit the half-precision rounding described above. The per-pixel affine refit and kernel correction require the dataset and member predictions on the relevant fitting splits. The kernel's posterior scale also requires its Gram matrix. The five-member archive does not contain the second experiment's completed FNO or added UNet.

Runtime. Table S9.1 reports wall times from the first experiment. Each task used six threads, with five tasks sharing an instance. Results are separated by host class: seeds 2, 4, 8 and 9 used the larger instance for the kernel stages. A factorization, solve and validation prediction at $n = 19000$ took 20–35 s there, compared with 42–113 s on the standard instances. The FNO's incomplete schedules are summarized by seconds per epoch from cumulative ten-epoch timers. Uninterrupted intervals took 467–625 s per epoch on standard instances and 348–410 s on the larger instance. Intervals affected by shared-host contention or suspension reached 10^4 s per epoch; the table therefore reports both minimum and median rates. Selected checkpoints were reached after 1.3–8.0 h at five seeds with uninterrupted timers, and 9.0–16.9 h at the remaining five. Full-test inference latency and peak memory were not recorded. Per-seed timings are provided in `runs/timing_harvest.json`.

Member costs. Table S9.2 gives training costs for the second experiment over ten seeds, with all schedules completed on one host. Thread counts differ across members because the host was shared with the rest of the experiment. We report thread-minutes, elapsed wall time multiplied by the configured thread count. This is allocated parallelism; CPU utilization was not measured, and the timings include contention on the shared host. The FNO uses about 7900 thread-minutes against 54 for the MSE MLP while improving single-member error by about 0.03 percentage points. The UNet uses about 115 times the MSE MLP's thread-minutes and has higher single-member error, although it improves the ensemble. Parameter counts alone do not explain these differences: the FNO has 6.45M parameters, the refiner 6.64M, and the UNet 4.38M. Convolution over the full field and dense prediction from

Table S9.1: Wall times for the first structural-mechanics experiment (all stages on CPU, six threads per task, five tasks sharing each instance). Network rows: the trainer's elapsed minutes from the first epoch to the final evaluation, as the range over seeds, with the scheduled epochs and the selected checkpoint epoch (range over the ten seeds). The FNO row gives seconds per epoch from ten-epoch timing intervals, minimum and median over all intervals (count in parentheses), since no seed completed the schedule. Kernel rows: the complete stage (fifteen grid cells and the refit) and its individual factorization-and-solve operations, as ranges, with the number of timed cells or folds in parentheses. The two columns are the two instance classes of the experiment, with the number of seeds in parentheses.

Stage	Epochs: scheduled / selected	Standard instances (6 seeds)	Larger instance (4 seeds)
Metric-loss MLP	400 / 39–79	129–158 min	26–35 min
MSE MLP	120 / 99–119	9.7–11.9 min	6.6–11.1 min
Refiner	100 / 99	9.9–13.1 min	7.9–33.7 min
FNO, per epoch	200 / 10–70	467 s min, 581 s median (88)	348 s min, 437 s median (56)
Kernel ridge, grid and refit	—	17.0–26.0 min	6.1–7.4 min
one factor-and-solve, $n = 19000$	—	42–113 s (90)	20–35 s (60)
out-of-fold fold, $n = 14250$	—	32–66 s (24)	13–22 s (16)

Table S9.2: Training costs of the second experiment on a shared CPU host with 20 physical cores and 40 hardware threads, every member trained to the end of its schedule; ranges over the ten seeds. Wall minutes are elapsed training times; thread-minutes is wall clock times the thread count (10 for the convolutional members, 10 to 16 for the MLPs); the kernel stage does not record a thread count. Test errors use reflection averaging for the neural members, as in Table 4. Per-seed values are provided in `campaign/collected/dgx/runtime_ledger.csv`.

Member	Parameters	Epochs	Wall (min)	Thread-min	Test error (%)
Kernel ridge, grid and refit	—	—	5.8–10.0	—	5.193–5.200
Metric-loss MLP	4.91M	400	8.8–19.2	132–192	4.809–4.858
MSE MLP	4.91M	120	2.6–5.8	39–58	4.703–4.719
Refiner	6.64M	100	2.6–6.1	41–61	4.721–4.742
FNO	6.45M	100	662–836	6618–8355	4.665–4.697
UNet	4.38M	100	573–689	5734–6892	4.683–4.793

the load vector have different computational costs; the table measures their implementations at the stated schedules.

Stacking. The five-member fitted-weight baseline uses least squares on flattened full-validation predictions, followed by clipping negative coefficients and normalizing the weights to sum to one (`campaign/ensemble_seeded5.py`). The six-member fitted-weight baseline uses full-validation mean relative error and softmax weights: `stack_correct.py` starts from uniform weights and takes 2000 Gaussian proposals in logit space, with initial step 0.5, decay factor 0.999 and random seed 0. The separate global-weight fit used to judge per-pixel stacking uses half the validation set, 600 proposals, initial step 0.3 and decay factor 0.997. The simplex minimizer of the second-moment matrix S (Proposition 6.1) is used only in the pool analyses. Per-pixel stacking fits an affine combination of member predictions at each grid point, with ridge parameter 10^{-3}. The weights are fitted on half the validation split and selected if their error on the other half is lower than that of global weights. The selected stack is then refitted on the full split.

Kernel stages. The kernels are Matérn-5/2 functions of standardized inputs. The residual correction searches the scale grid $\{1, 2, 4\}\times$ the median pairwise distance (estimated on 2000 fitting rows) and the nugget grid $\{10^{-6}, 10^{-5}, 10^{-3}\}$ (scaled by n); the low-data chain of Table 3 uses $\{0.5, 1, 2, 4\}$ with nuggets $\{10^{-7}, 10^{-5}, 10^{-3}\}$, with the median computed on all 1000 fitting rows. Candidate fits use $\min(8000, n)$ fitting rows, where n is the fitting-set size: 8000 rows in the high-data protocol and all 1000 rows in the low-data protocol. Hyperparameters are selected on validation, then refitted on all fitting rows in double precision. The pure kernel baseline (Table 2) uses the grid $\{0.5, 0.75, 1, 1.5, 2\}\times$median and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$ directly at $n = 19000$. The feature-space stage concatenates and standardizes penultimate activations, then uses the same kernel family and tuning protocol. These features are spatially averaged pre-projection channels for the FNO (128 coordinates), mean-pooled encoder tokens for the transformer (192), and the residual-trunk output for the MLP (1024).

Low-data correction sequence. The low-data sequence uses one trained MLP as its initial mean. The candidate predictors are the neural mean, the mean plus an input-space correction, that predictor plus a feature-space correction, and the resulting predictor plus a second input-space correction. Each correction is fitted to the remaining training residuals. The final predictor is the candidate with the lowest error on all 250 validation rows.

Low-data multiscale kernel. The kernel baseline in Table 3 uses the additive residual construction in `campaign/krr_multiscale_lowdata.py`. Inputs are standardized by the coordinatewise mean and standard deviation of the $n = 1000$ fitting rows. Initialize $g_0(x) = \bar{y}$, the fitting-target mean. At stage t, each candidate fits the current residual matrix $R_t = Y - g_{t-1}(X)$ and proposes

$$g_t(x) = g_{t-1}(x) + k_{\ell_t}(x, X)(K_{\ell_t} + n\lambda_t I)^{-1} R_t.$$

The Matérn-5/2 length scale is selected from $\{0.25, 0.5, 1, 2, 3\}d_{\mathrm{med}}$, with

$$d_{\mathrm{med}} = \left(\mathrm{median}_{i<j} \|x_i - x_j\|_2^2\right)^{1/2}$$

computed on the standardized fitting inputs. The nugget grid is $\{10^{-8}, 10^{-6}, 10^{-4}\}$. At each stage the candidate with the lowest mean relative-L^2 error on the fixed 250-row validation split is selected. Its increment is retained with coefficient one only if that error improves by more than 10^{-5} in relative-error units. Otherwise the procedure stops. At most four stages are retained, and all fits use the same 1000 fitting rows. The procedure is deterministic for this fixed split.

OCO-2 network and feature heads. The implementation in `campaign/jpl_seeded.py` uses a width-384 residual MLP. A linear input map and SiLU activation precede three residual layers $h \mapsto h + \mathrm{SiLU}(Wh + b)$ and a linear output map to the 40 standardized PCA coordinates. The final 384-dimensional hidden vector supplies the frozen features for the kernel and ridge readouts. Training uses AdamW with learning rate 10^{-3}, weight decay

10^{-5}, batch size 512 and a cosine schedule to 10^{-6} over 250 epochs. Validation is evaluated every 25 epochs and at the final epoch; the checkpoint minimizing the corresponding training metric on validation is retained. The three losses are mean relative error in standardized PCA coordinates, mean relative error after diagonal scaling by the PCA coefficient scales, and mean relative error after the retained-PCA radiance reconstruction. The additional 750-epoch run uses the same architecture and optimizer, with the cosine schedule extended to that budget.

For each OCO-2 kernel head, features are standardized by fitting-set moments. The Matérn-5/2 scale grid is $\{0.5, 1, 2, 4\}d_{\text{med}}$ and the nugget grid is $\{10^{-8}, 10^{-6}, 10^{-4}\}$. A seeded 6000-row fitting subsample is used both to estimate d_{med} and to fit candidates, which are selected on the full validation split using the associated loss. The selected pair is refitted on all 18000 fitting rows. Raw-input kernels use the same grid, subsample size and refitting protocol. The ridge control includes an intercept and selects λ from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$ on validation, with normal equations regularized by $n\lambda I$.

OCO-2 input sensitivity. Let X_s contain the fitting inputs standardized by their coordinatewise fitting means and standard deviations, and let Z contain the released standardized PCA targets. The sensitivity estimator in `campaign/jpl_seeded.py` is the least-squares matrix

$$A = X_s^\dagger Z, \qquad a_j = \|A_{j,:}\|_2, \qquad w_j = \frac{a_j}{d^{-1}\sum_{k=1}^d a_k}.$$

The adapted kernel uses distances between $\operatorname{diag}(w)x_s$. The weights are applied after standardization, so subsequent input normalization does not remove them. The estimator is refitted using only the available fitting rows at each training size. The rank summaries in `campaign/scaling_seeded.py` instead use $B = X_{\text{tr}}^\dagger Z$ on all 18000 fitting rows, where X_{tr} contains the released input coordinates without the additional fitting-set centering and standardization. For the singular values σ_j of B, the 99% count is the smallest r satisfying $\sum_{j\le r}\sigma_j^2 \ge 0.99\sum_j \sigma_j^2$, and the reported participation ratio is $(\sum_j \sigma_j)^2/\sum_j \sigma_j^2$.

Separate O2 learned-metric experiment. The experiment in `learnable_kernel_gpu .py` compares an isotropic metric, a learned diagonal metric, and $M = L^\top L + \operatorname{diag}(\exp(2b))$, with $L \in \mathbb{R}^{6\times d}$. The diagonal model has d metric parameters and the low-rank model $7d$. After a fixed seed-0 permutation, kernel fitting uses 6000 pool rows and validation uses the next 2000. Metric learning uses 400 Adam steps at learning rate 0.05, with 256 fitting rows sampled with replacement per step. For a batch b and its first half c, let K_b and K_c be the Matérn Gram matrices and let Z_b and Z_c contain their reduced targets. The regularized kernel-flows objective is

$$\rho = \max\left\{10^{-6},\, 1 - \frac{\operatorname{tr}\big[Z_c^\top (K_c + \eta I)^{-1} Z_c\big]}{\operatorname{tr}\big[Z_b^\top (K_b + \eta I)^{-1} Z_b\big] + 10^{-30}}\right\}, \qquad \eta = \exp(\delta) + 10^{-8}.$$

This is the regularized half-batch criterion of Owhadi and Yoo (2019), with the quadratic forms summed over outputs. The parameter δ is learned jointly with the metric. With the learned metric fixed, validation selects a multiplier of its feature map from $\{0.1, 0.25, 0.5, 1, 2, 4, 8\}$ and a nugget from $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The final Matérn-5/2 solve uses the same 6000 fitting rows.

The raw-state experiment has $d = 20$, so the low-rank term adds 120 parameters. The feature experiment uses a separate width-256 MLP with two residual SiLU blocks, trained by standardized-target MSE for 600 epochs on the 20000-row pool before the kernel split. Its $d = 256$ feature map gives 1536 additional low-rank parameters. These experiments use different training budgets and representations from Table 6; in the feature experiment, the kernel-validation targets have already contributed to neural training. They provide a separate comparison of metric choices within those fitted representations.

Uncertainty. The posterior standard deviation in (4) is computed by triangular solves using the Cholesky factor of $K + n\lambda I$. Conformal calibration uses a random 1000-sample subset of the public test block and evaluation uses the disjoint remainder. Model-fitting optimizers use training and validation. The marginal coverage guarantee of Proposition S3.1 requires the predictor and scale to be fixed independently of exchangeable calibration and evaluation observations. The exact Beta reference law additionally assumes i.i.d. continuous scores. The miscoverage levels $\alpha = 0.10$ and 0.05 have $k \leq m$; for smaller miscoverage with $k = m + 1$, the mathematical quantile is $+\infty$, not the largest finite calibration score.

S10 Numerical evaluation of the theoretical identities and bounds

Numerical experiments evaluate the identities and inequalities on synthetic designs and on the benchmark residual statistics. The sixty-predictor second-moment matrix gives an empirical RMS lower bound of 4.813611% and a convex optimum of 4.876133%.

The second-moment identity and validation dispersion factors 0.938 and 0.938 use the same samples as their respective quadratic-form calculations. The normalized five-member value 0.980 ± 0.003 is the held-out risk of fitted global weights. The six-member value 0.972 ± 0.005 is the validation simplex minimum.

The central 95% conformal reference intervals at $m = 1000, N = 19000$ are $[88.0\%, 91.8\%]$ and $[93.5\%, 96.3\%]$ at nominal 90% and 95%. Their Beta–binomial interpretation is stated in Remark S3.2.

Residual-mixture identities. In the exact block-correlated model, the quadratic-form identity in the proof of Theorem 6.6 has maximum absolute discrepancy 3×10^{-18} over two thousand random convex weight vectors. On a perturbed pool and on the sixty-predictor matrix, all twenty thousand sampled convex mixtures satisfy the bound obtained from the pool's minima. Replacing block minima by block means gives the uniform mixture's value. The signed optimum of Corollary 6.5 agrees with its closed form and is no larger than any sampled convex-mixture risk.

Local sensitivity. The two derivatives in Proposition 6.7 agree with central differences to 4×10^{-9} over two thousand interior points. Both are positive at the sampled points away from perfect anticorrelation; along the stated direction, the change in optimal risk is second order in the step.

Kernel bounds. The bound in Lemma S1.7 holds on fifteen hundred random Matérn designs and evaluation points, with a largest ratio to the bound of 0.87; the cosine-kernel construction attains equality to 7×10^{-15}. The maximum discrepancy in the identity of Theorem 6.10 is 4×10^{-14}, and $P_0 \leq \widetilde{P}_\lambda \leq P_\lambda$ on every sampled design. The extremizing residual

Table S11.1: Target-centering comparison on the fixed 20000-case mechanics test block, ten paired seeds. The baseline pipeline is shown as a reference; both centering choices use the same training schedule and CPU-thread setting.

Centering convention	Mean relative error (%)
Original retained pipeline	4.5460 ± 0.0029
Pooled centering, paired rerun	4.5459 ± 0.0029
Fold-local centering, paired rerun	4.5459 ± 0.0029
Fold-local minus pooled (percentage points)	0.000011 ± 0.000110

in the proof of Theorem 6.9 attains its bound to 10^{-14}. The numerical implementation is provided in `campaign/verify_floor_and_certificate.py`.

S11 Implementation sensitivity and prediction reconstruction

These experiments address three implementation choices: target centering in out-of-fold kernel fits, consistency of correction labels with the query-time mean, and the extent of the OCO-2 kernel search. Seeds and search grids were specified before the experiments. The $\pm$ values below are sample standard deviations over the stated seeds.

S11.1 Paired target-centering comparison

For each mechanics seed, the four kernel folds contain 14250 fitting rows and 4750 held-out rows. The pooled arm subtracts the mean target over all 19000 training rows. The fold-local arm instead subtracts the mean over that fold's fitting rows. Input standardization, kernel scale, nugget 10^{-6}, and fold assignment remain fixed. In particular, "fold-local" specifies target centering only; input standardization uses the full training block.

The two predictions can be calculated from the same factorization. If $A = K_{FF} + |F|\lambda I$, $k_H = k(X_H, X_F)$, and μ_F, μ_P are the fold and pooled target means, their difference on held-out rows is

$$(\mathbf{1} - k_H A^{-1}\mathbf{1})(\mu_F - \mu_P)^\top.$$

At eight prespecified output coordinates, a separate right-hand-side solve gives the same prediction difference. The pooled predictions also reproduce the baseline out-of-fold field.

Each arm retrains the refiner for the same 100 epochs, seed, and eight-thread CPU setting, then repeats global stacking, the held-out decision about per-pixel weights, full-validation refitting, kernel selection, and calibration. The other four neural predictors and the full-training kernel predictor are held fixed.

S11.2 Correction-label consistency with a fixed estimator

For each six-member pipeline, the training-row mean is reconstructed and its query-time counterpart is evaluated on those same rows. The selected kernel, its hyperparameters, and the fitted stack weights remain fixed. Two sets of coefficients are computed: one from the mixed residual labels, the other from $G(X) - m_{\mathrm{full}}(X)$. The difference of their predictions

Table S11.2: Calibration under the two centering choices, ten seeds. The same 1000 calibration cases and 19000 evaluation cases are used in each paired arm. Radii are absolute Euclidean norms on the output grid. "Constant" uses scale one; "disagreement" uses the ensemble spread. These are measurements on the public test block; the independence conditions are stated in Supplement S3.

		Coverage (%)		Mean radius	
Target	Scale	Pooled	Fold-local	Pooled	Fold-local
90%	Constant	90.40 ± 0.80	90.38 ± 0.77	351.0 ± 4.3	350.9 ± 4.2
90%	Disagreement	90.05 ± 0.40	90.09 ± 0.36	405.8 ± 3.7	406.2 ± 3.9
95%	Constant	95.38 ± 0.54	95.39 ± 0.55	394.3 ± 6.3	394.4 ± 6.4
95%	Disagreement	95.01 ± 0.77	94.98 ± 0.72	466.5 ± 12.6	465.9 ± 11.6

Table S11.3: Correction-label sensitivity, ten mechanics pipelines. No neural training or stack-weight fitting is repeated. The kernel is the one selected by the baseline pipeline.

Fixed original mean and kernel	Mean relative error (%)
Mean without correction	4.5666 ± 0.0038
Original residual labels	4.5460 ± 0.0029
Inference-consistent residual labels	4.5460 ± 0.0029
Consistent correction or mean, validation-selected	4.5460 ± 0.0029
Consistent minus original (percentage points)	0.000007 ± 0.000029
Mean propagated mismatch, relative norm (%)	0.0148 ± 0.0035

is exactly the propagated term in (5). The coefficient difference agrees with this identity, and the original coefficients reproduce the validation and test predictions. The change in the error norm is bounded by the propagated mismatch norm on every evaluated case, as required by the reverse-triangle inequality.

Table S11.3 also reports a two-candidate decision: retain the uncorrected mean or use its inference-consistent correction, according to validation error. This decision does not retune the kernel grid. The separate correction row gives its error regardless of the validation selection. The largest propagated relative norm over all test cases and ten seeds is 0.2789%; this maximum is a finite-sample diagnostic.

S11.3 A wider OCO-2 kernel search

The grid comparison uses seeds 0, 1, and 2 in each of the three bands. At each band-seed, the three networks are trained for 250 epochs with width 384, then their predictions and features are held fixed for both searches. Both grids use the same training/validation split, the same 6000-row tuning subsample, the same objective for each head, and a refit at the selected hyperparameters on all 18000 training rows.

The baseline grid has scale multipliers $\{0.5, 1, 2, 4\}$ and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$, for 12 cells per kernel. The expanded grid has multipliers $\{0.25, 0.5, 1, 2, 4, 8, 16\}$ and all eight

Table S11.4: O2 grid comparison, three paired seeds. The errors are means and sample standard deviations over seeds. Of the 15 selected kernels (five per seed), 13 meet a boundary of the baseline grid and 5 meet a boundary of the expanded grid. The respective failed-cell counts are 0 and 0.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	16.889 ± 0.000	16.889 ± 0.000	0.0448 ± 0.0000	0.0448 ± 0.0000
Raw input	40.529 ± 0.089	40.529 ± 0.089	0.2353 ± 0.0047	0.2353 ± 0.0047
ARD input	30.019 ± 0.035	29.062 ± 0.165	0.2706 ± 0.1030	0.1519 ± 0.0007
Flat network	4.440 ± 0.087	4.440 ± 0.087	0.1893 ± 0.0158	0.1893 ± 0.0158
Flat linear readout	4.774 ± 0.094	4.774 ± 0.094	0.2176 ± 0.0109	0.2176 ± 0.0109
Flat kernel head	4.104 ± 0.074	4.331 ± 0.074	0.0821 ± 0.0092	0.1260 ± 0.0203
Weighted network	49.247 ± 0.008	49.247 ± 0.008	0.0448 ± 0.0036	0.0448 ± 0.0036
Weighted linear readout	46.419 ± 0.328	46.419 ± 0.328	0.0539 ± 0.0009	0.0539 ± 0.0009
Weighted kernel head	21.130 ± 0.281	22.308 ± 0.793	0.0306 ± 0.0010	0.0306 ± 0.0013
Radiance network	59.048 ± 0.406	59.048 ± 0.406	0.0534 ± 0.0047	0.0534 ± 0.0047
Radiance linear readout	44.689 ± 0.909	44.689 ± 0.909	0.0562 ± 0.0043	0.0562 ± 0.0043
Radiance kernel head	20.141 ± 0.361	26.991 ± 0.283	0.0351 ± 0.0008	0.0500 ± 0.0093
Coordinate combination	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012
Combination with linear readouts	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012

powers $10^{-10}, \ldots, 10^{-3}$, for 56 cells. Scales multiply the median distance in the representation being fitted. The isotropic raw-input kernel, ARD-input kernel, and three feature heads all receive these same candidate grids. A Cholesky failure or nonfinite score is recorded as a failed cell; only finite successful validation scores can be selected. Boundary selections and failed cells are counted in the table captions.

The frozen linear readouts retain the same ridge search, and the source emulator's predictions are unchanged. The coordinate selector is refitted on validation for each grid because its candidate kernel heads change. The tables report every candidate, including the additional selector that also admits the linear readouts. Both errors refer to the retained forty-component representation, as in the main OCO-2 study.

S11.4 Reconstruction from saved prediction fields

The mechanics reconstruction subtracts the saved reference fields in float64, recomputes the errors of all sixty retained members, and accumulates the 60×60 evaluation second-moment matrix in blocks of 200 cases. It uses the same fixed 1000/19000 fitting/evaluation split. The largest entrywise difference from the retained matrix is 4.32×10^{-8}. An active-set solve, checked by a convex first-order lower bound, changes the optimal RMS relative error by 1.88×10^{-7} percentage points. The block lower bound and the convex optimum agree with the retained analysis to four decimal places when expressed as percentages.

The mechanics accuracy uses the ordinary Euclidean norm on the 41×41 output grid. A comparison with spatial quadrature applies tensor-product trapezoidal weights: one in the interior, one half on edges, and one quarter at corners. The common grid-spacing factor cancels in the relative error. For the ten corrected pipelines, the plain-grid error is $4.5460 \pm 0.0029\%$, while the trapezoidal-grid error is $4.3666 \pm 0.0030\%$. Nested one-

Table S11.5: WCO2 grid comparison, three paired seeds, with the same protocol as Table S11.4. Boundary selections: 9 and 7 out of 15; failed cells: 0 and 0, for the baseline and expanded grids respectively.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	24.060 ± 0.000	24.060 ± 0.000	0.0599 ± 0.0000	0.0599 ± 0.0000
Raw input	58.625 ± 0.202	58.625 ± 0.202	0.4431 ± 0.0063	0.4431 ± 0.0063
ARD input	52.454 ± 0.071	53.505 ± 0.073	0.2231 ± 0.0014	0.7523 ± 0.0106
Flat network	16.419 ± 0.070	16.419 ± 0.070	0.2268 ± 0.0049	0.2268 ± 0.0049
Flat linear readout	16.491 ± 0.061	16.491 ± 0.061	0.2502 ± 0.0136	0.2502 ± 0.0136
Flat kernel head	16.270 ± 0.067	16.395 ± 0.073	0.1154 ± 0.0012	0.1770 ± 0.0046
Weighted network	62.501 ± 0.392	62.501 ± 0.392	0.0742 ± 0.0096	0.0742 ± 0.0096
Weighted linear readout	61.294 ± 0.319	61.294 ± 0.319	0.0842 ± 0.0108	0.0842 ± 0.0108
Weighted kernel head	48.826 ± 0.472	50.305 ± 1.924	0.0544 ± 0.0070	0.0626 ± 0.0179
Radiance network	67.505 ± 0.410	67.505 ± 0.410	0.0584 ± 0.0063	0.0584 ± 0.0063
Radiance linear readout	61.206 ± 1.096	61.206 ± 1.096	0.0626 ± 0.0033	0.0626 ± 0.0033
Radiance kernel head	48.247 ± 0.576	50.342 ± 0.397	0.0453 ± 0.0059	0.0558 ± 0.0073
Coordinate combination	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056
Combination with linear readouts	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056

Table S11.6: SCO2 grid comparison, three paired seeds, with the same protocol as Table S11.4. Boundary selections: 15 and 3 out of 15; failed cells: 0 and 0, for the baseline and expanded grids respectively.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	16.145 ± 0.000	16.145 ± 0.000	0.1147 ± 0.0000	0.1147 ± 0.0000
Raw input	41.012 ± 0.090	43.708 ± 0.065	0.5229 ± 0.0081	0.5738 ± 0.0054
ARD input	28.717 ± 0.021	28.717 ± 0.021	0.6023 ± 0.0097	0.6023 ± 0.0097
Flat network	8.236 ± 0.021	8.236 ± 0.021	0.2060 ± 0.0209	0.2060 ± 0.0209
Flat linear readout	8.340 ± 0.024	8.340 ± 0.024	0.2436 ± 0.0528	0.2436 ± 0.0528
Flat kernel head	8.092 ± 0.018	8.108 ± 0.025	0.1094 ± 0.0051	0.1142 ± 0.0093
Weighted network	36.028 ± 0.697	36.028 ± 0.697	0.0740 ± 0.0022	0.0740 ± 0.0022
Weighted linear readout	39.605 ± 0.369	39.605 ± 0.369	0.0858 ± 0.0041	0.0858 ± 0.0041
Weighted kernel head	14.908 ± 0.219	15.347 ± 0.194	0.0614 ± 0.0036	0.0609 ± 0.0023
Radiance network	47.386 ± 1.149	47.386 ± 1.149	0.0960 ± 0.0088	0.0960 ± 0.0088
Radiance linear readout	41.112 ± 1.917	41.112 ± 1.917	0.0994 ± 0.0080	0.0994 ± 0.0080
Radiance kernel head	15.506 ± 0.869	19.022 ± 2.830	0.0800 ± 0.0090	0.0904 ± 0.0174
Coordinate combination	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033
Combination with linear readouts	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033

dimensional quadrature reproduces the weighted calculation. The predictors are fixed and the external comparators are not rescored; the main comparison continues to use the stated plain-grid metric.

S11.5 Recomputation of errors and calibration

Summary statistics, calibration quantiles and validation selections are recomputed from the per-case predictions. For the twenty target-centering fits, relative and absolute errors are calculated directly from the stored fields. The six-member disagreement scale is recomputed using the identity

$$\frac{1}{M}\sum_{a=1}^{M}(x_a - \bar{x})^2 = \frac{1}{M^2}\sum_{a<b}(x_a - x_b)^2$$

at every output coordinate, followed by the same coordinate average of the standard deviations. For OCO-2, scores for all fourteen predictors are recomputed under both grids in all nine band–seed combinations. Radiance errors use a spectral Gram quadratic form, with selected cases also evaluated by direct full-spectrum summation. Coordinate selectors are reconstructed from the validation predictions.

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